Chapter topics:

The transition elements, Electronic configuration, special properties
Nomenclature
Werner's theory
Electronic interpretation of coordination
Pauling's Valence Bond Theory
Isomerism stereochemistry of 4- and 6- coordinate complexes
Chelate complexes, examples
Stabilization of unusual oxidation states
Application of coordination compounds

Tetramethyldiarsine a.k.a Cacodyl, the first organometallic compound deliberately synthesized. **French chemist Louis Claude Cadet de Gassicourt** isolated the first organometallic compound, tetramethyldiarsine a.k.a. cacodyl, in **1757** by accident.

Image result for was zeise's salt the first organometallic compound History. Zeise's salt was one of the first organometallic compounds to be reported. It was discovered by William Christopher Zeise, a professor at the University of Copenhagen, who prepared this compound in **1830** while investigating the reaction of $PtCl_4$ with boiling ethanol.

Structural bio-organometallic chemistry has its origin with the English chemist Dorothy Crowfoot-Hodgkin who **established the X-ray crystal structure of vitamin B12 coenzyme between 1953 and 1961** using a primitive computer, resulting in her awarding the Nobel Prize in 1964.

The modern theory of coordination chemistry is based largely on the work of Alfred Werner (1866–1919; Nobel Prize in Chemistry in 1913). In a series of careful experiments carried out in the late 1880s and early 1890s, he examined the properties of several series of metal halide complexes with ammonia. For example, five different "adducts" of ammonia with $PtCl_4$ were known at the time: $PtCl_4 \cdot nNH_3$ (n = 2–6). Some of Werner's original data on these compounds are shown in Table 1. The electrical conductivity of aqueous solutions of these compounds was roughly proportional to the number of ions formed per mole, while the number of chloride ions that could be precipitated as AgCl after adding $Ag^+(aq)$ was a measure of the number of "free" chloride ions present. For example, Werner's data on $PtCl_4 \cdot 6NH_3$ in Table 1

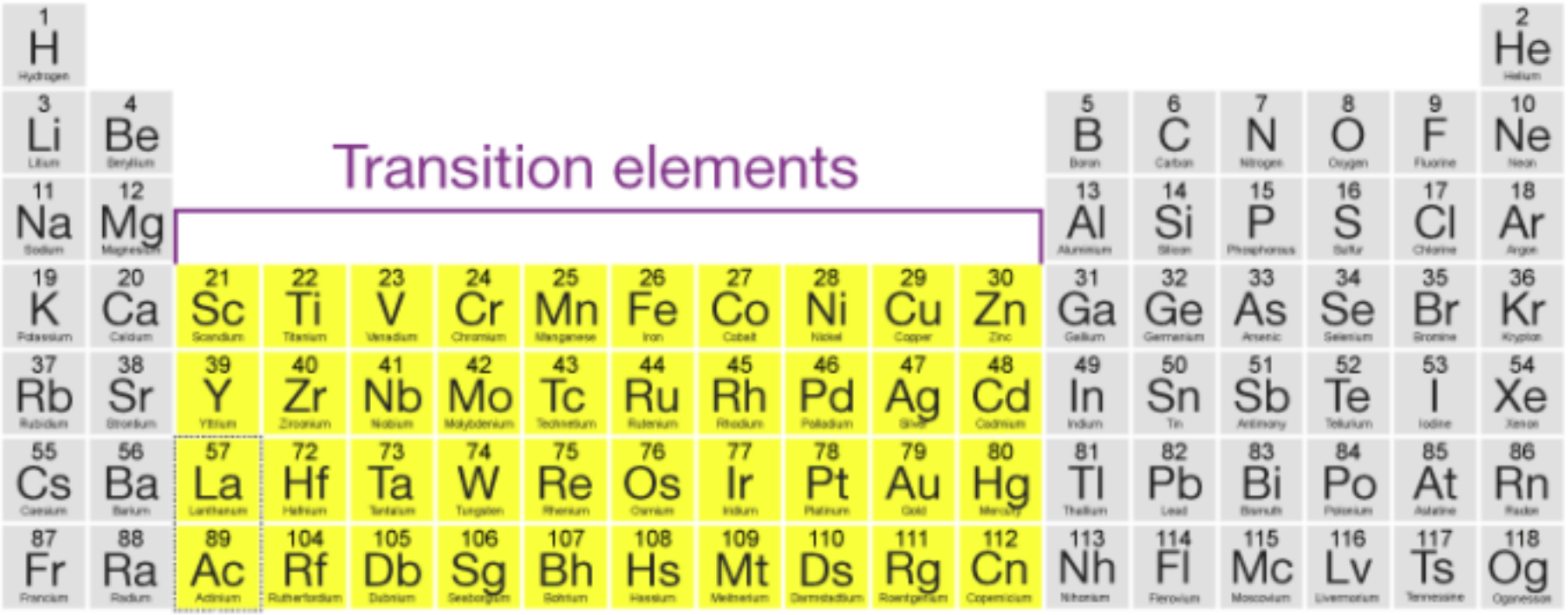
showed that all the chloride ions were present as free chloride. In contrast, $PtCl_4 \cdot 2NH_3$ was a neutral molecule that contained no free chloride ions.

Alfred Werner (1866–1919)

Werner, the son of a factory worker, was born in Alsace. He developed an interest in chemistry at an early age, and he did his first independent research experiments at age 18. While doing his military service in southern Germany, he attended a series of chemistry lectures, and he subsequently received his PhD at the University of Zurich in Switzerland, where he was appointed professor of chemistry at age 29. He won the Nobel Prize in Chemistry in 1913 for his work on coordination compounds, which he performed as a graduate student and first presented at age 26. Apparently, Werner was so obsessed with solving the riddle of the structure of coordination compounds that his brain continued to work on the problem even while he was asleep. In 1891, when he was only 25, he woke up in the middle of the night and, in only a few hours, had laid the foundation for modern coordination chemistry.

TRANSITION ELEMENTS

BYJU'S The Learning App

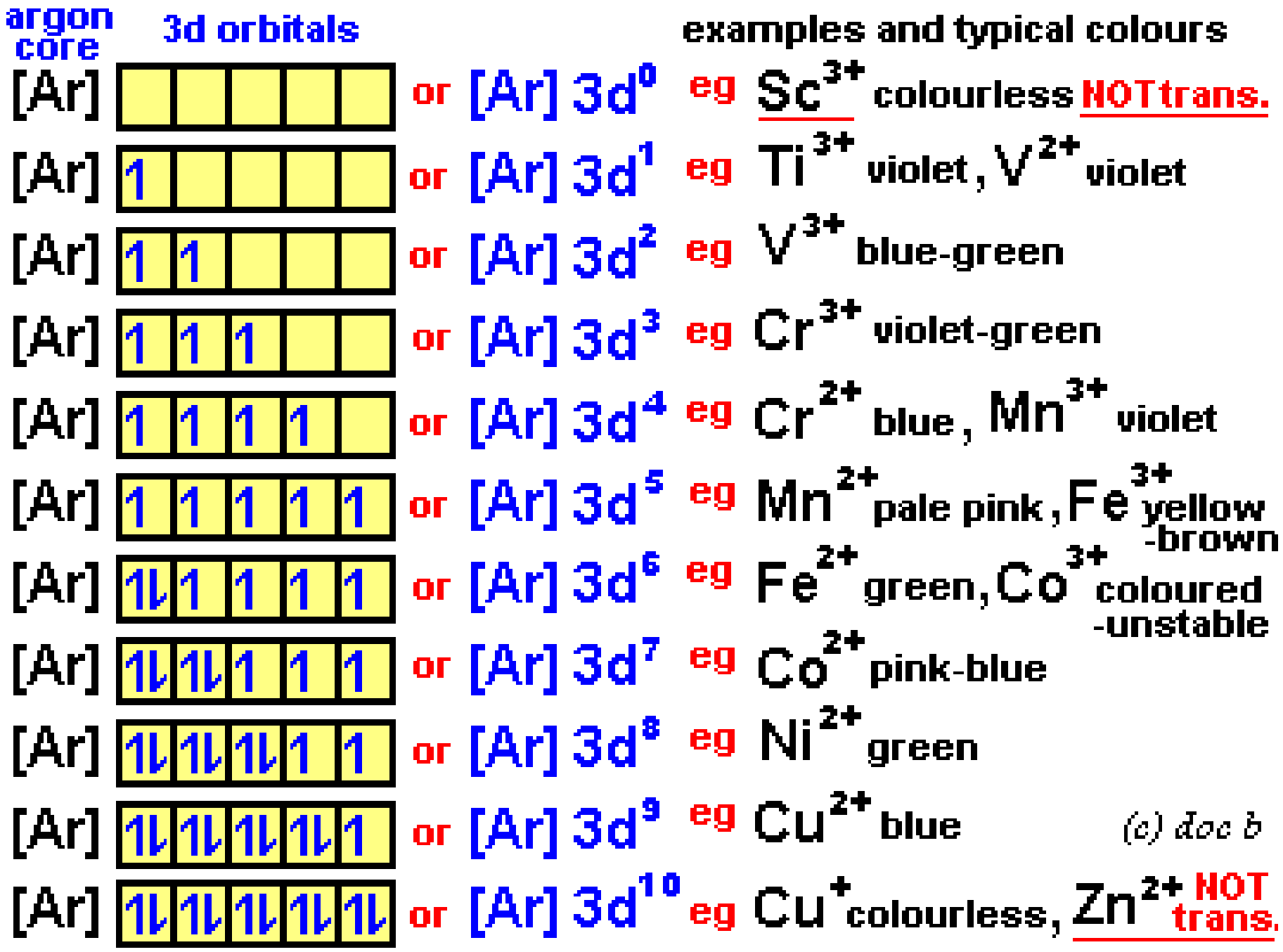


© Byjus.com

Answer

Element	General outer electronic configuration
s–block(alkali metals)	ns^{1-2}, where n = 2 – 7
p–block(metals & non metals)	ns^2np^{1-6}, where n = 2 – 6
d–block(transition elements)	$(n-1)\,d^{1-10}\,ns^{0-2}$, where n = 4 – 7
f–block(inner transition elements)	$(n-2)f^{1-14}(n-1)d^{0-10}ns^2$, where n = 6 – 7

Element	Symbol	Z	Electronic Configuration
Scandium	Sc	21	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^1\,4s^2$
Titanium	Ti	22	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^2\,4s^2$
Vanadium	V	23	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^3\,4s^2$
Chromium	Cr	24	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5\,4s^1$
Manganese	Mn	25	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5\,4s^2$
Iron	Fe	26	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^6\,4s^2$
Cobalt	Co	27	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^7\,4s^2$
Nickel	Ni	28	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^8\,4s^2$
Copper	Cu	29	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^1$
Zinc	Zn	30	$1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^2$


argon core
3d orbitals
examples and typical colours
[Ar]
or [Ar] 3d0 eg Sc3+ colourless NOT trans.
[Ar] 1
or [Ar] 3d1 eg Ti3+ violet, V2+ violet
[Ar] 1 1
or [Ar] 3d2 eg V3+ blue-green
[Ar] 1 1 1
or [Ar] 3d3 eg Cr3+ violet-green
[Ar] 1 1 1 1
or [Ar] 3d4 eg Cr2+ blue, Mn3+ violet
[Ar] 1 1 1 1 1
or [Ar] 3d5 eg Mn2+ pale pink, Fe3+ yellow -brown
[Ar] 1↓ 1 1 1 1
or [Ar] 3d6 eg Fe2+ green, Co3+ coloured -unstable
[Ar] 1↓ 1↓ 1 1 1
or [Ar] 3d7 eg Co2+ pink-blue
[Ar] 1↓ 1↓ 1↓ 1 1
or [Ar] 3d8 eg Ni2+ green
[Ar] 1↓ 1↓ 1↓ 1↓ 1
or [Ar] 3d9 eg Cu2+ blue
(c) doc b
[Ar] 1↓ 1↓ 1↓ 1↓ 1↓
or [Ar] 3d10 eg Cu+ colourless, Zn2+ NOT trans.

Transition metals
they are good conductors of heat and electricity.
they can be hammered or bent into shape easily.
they have high melting points (but mercury is a liquid at room temperature)
they are usually hard and tough.
they have high densities.

What is unique about transition metals?
The transition elements are unique in that they **can have an incomplete inner subshell allowing valence electrons in a shell other than the outer shell**. Other elements only have valence electrons in their outer shell. This allows transition metals to form several different oxidation states.

Transition metals have a higher charge-to-radius ratio than **main group metals**, and are the only **metals** known to produce paramagnetic compounds. **Transition metals** are used as catalysts in reactions more often than **main group metals**.

(i) They are all metals and as such are lustrous and deformable and have high electrical and thermal conductivities. In addition, their melting and boiling points tend to be high and they are generally hard and strong.

(ii) Most of them display numerous oxidation states which vary by steps of 1 rather than 2 as is usually the case with those main-group elements which exhibit more than one oxidation state.

(iii) They have an unparalleled propensity for forming coordination compounds with Lewis bases.

A coordination compound, or complex, is formed when a Lewis base (ligand)[1] is attached to a Lewis acid (acceptor) by means of a "lone-pair" of electrons. Where the ligand is composed of a number of atoms, the one which is directly attached to the acceptor is called the "donor atom". This type of bonding has already been discussed (p. 198) and is exemplified by the addition compounds formed by the trihalides of the elements of Group 13 (p. 237); it is also the basis of much of the chemistry of the transition elements. The precise nature of the bond between a transition metal ion and a ligand varies enormously and the term "donor atom" is often used in situations where its literal meaning should not be assumed. Although inevitably the line of demarcation is rather ill-defined, it is conventional to distinguish two extremes. On the one hand, are those cases in which the bond may be considered profitably as a single σ bond, or even a purely electrostatic interaction, and in which the metal has an oxidation state of +2 or higher. On the other hand, are those cases where the bonding is multiple, the ligand acting simultaneously as both a σ donor and a π acceptor (p. 922) and in which the metal usually has a formal oxidation state of +1 or less, though the significance of such values is often unclear. Compounds of the former type are commonly described as "classical" or "Werner" complexes since it was through the investigation of such materials that A. Werner in the period 1893–1913 laid the foundations of coordination chemistry[2] (see also p. 912). Compounds of the latter type are exemplified by the carbonyls and other organometallic compounds.

N. N. Greenwood

Nomenclature

The International Union of Pure and Applied Chemistry (IUPAC) publication *Nomenclature of Inorganic Chemistry* (1989). Blackwell Scientific Publishers, contains the rules for systematic naming of coordination compounds. The basic rules are summarized here .

Nomenclature Rules

1. The cation comes first, followed by the anion.

   Examples: diamminesilver(I) chloride, $[Ag(NH_3)_2]Cl$

   potassium hexacyanoferrate(III), $K_3[Fe(CN)_6]$

   cation

   cation

2. The inner coordination sphere is enclosed in square brackets. Although the metal is provided first within the brackets, the ligands within the coordination sphere are written before the metal in the formula name.

   Examples: tetraamminecopper(II) sulfate, $[Cu(NH_3)_4]SO_4$

   hexaamminecobalt(III) chloride, $[Co(NH_3)_6]Cl_3$

3. When writing the formula of complexes, ligands are named before the metal. The coordinated groups are listed in the order: negative ligands, neutral ligands, positive ligands (and alphabetically according to the first symbol within each group).

(a) The names of negative ligands end in -o, for example:

F^-	fluoro	H^-	hydrido	HS^-	mercapto
Cl^-	chloro	OH^-	hydroxo	S^{2-}	thio
Br^-	bromo	O^{2-}	oxo	CN^-	cyano
I^-	iodo	O_2^{2-}	peroxo	NO_2^-	nitro

(b) Neutral groups have no special endings. Examples include NH_3 ammine, H_2O aqua, CO carbonyl and NO nitrosyl. The ligands N_2 and O_2 are called dinitrogen and dioxygen. Organic ligands are usually given their common names, for example phenyl, methyl, ethylenediamine, pyridine, triphenylphosphine.

(c) Positive groups end in -ium, e.g. ~~$NH_2—NH_2$~~ $[H_2N\text{-}NH_3]^+$ hydrazinium.

TABLE 2 Classic Monodentate Ligands

Common Name	IUPAC Name	Formula
hydrido	hydrido	H^-
fluoro	fluoro	F^-
chloro	chloro	Cl^-
bromo	bromo	Br^-
iodo	iodo	I^-
nitrido	nitrido	N^{3-}
azido	azido	N_3^-
oxo	oxido	O^{2-}
cyano	cyano	CN^-
thiocyano	thiocyanato-*S* (S-bonded)	SCN^-
isothiocyano	thiocyanato-*N* (N-bonded)	NCS^-

(continues)

TABLE 2 Classic Monodentate Ligands *(cont.)*

Common Name	IUPAC Name	Formula
hydroxo	hydroxo	OH^-
aqua	aqua	H_2O
carbonyl	carbonyl	CO
thiocarbonyl	thiocarbonyl	CS
nitrosyl	nitrosyl	NO^+
nitro	nitrito-*N* (N-bonded)	NO_2^-
nitrito	nitrito-*O* (O-bonded)	ONO^-
methyl isocyanide	methylisocyanide	CH_3NC
phosphine	phosphane	PR_3
pyridine	pyridine (abbrev. py)	C_5H_5N
ammine	ammine	NH_3
methylamine	methylamine	$MeNH_2$
amido	azanido	NH_2^-
imido	azanediido	NH^{2-}

TABLE 3 Chelating Amines

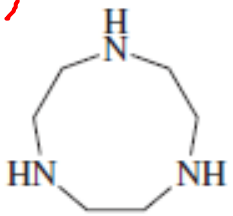
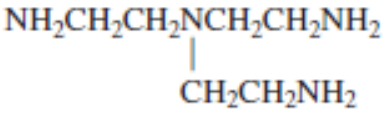
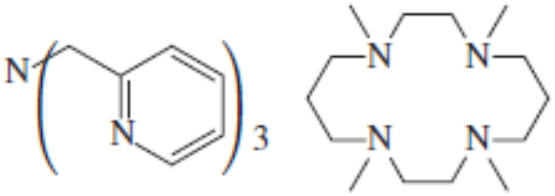
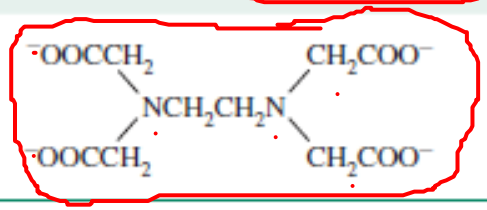
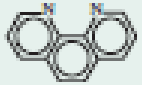
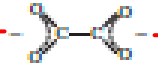
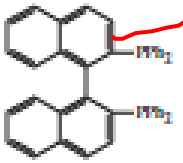
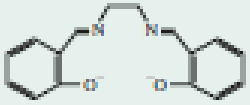
Chelating Points	Common Name	IUPAC Name	Abbrev.	Formula
bidentate	ethylenediamine	1,2-ethanediamine	en	$NH_2CH_2CH_2NH_2$
tridentate	diethylenetriamine	1,4,7-triazaheptane	dien	$NH_2CH_2CH_2NHCH_2CH_2NH_2$
		1,3,7-triazacyclononane	tacn	
tetradentate	triethylenetetraamine	1,4,7,10-tetraazadecane	trien	$NH_2CH_2CH_2NHCH_2CH_2NHCH_2CH_2NH_2$
	β, β', β''-triaminotriethylamine	β, β', β''-tris(2-aminoethyl)amine	tren	
	tetramethylcyclam	1,4,8,11-tetramethyl-1,4,8,11-tetraazacyclotetradecane	TMC	
	tris(2-pyridylmethyl) amine	tris(2-pyridylmethyl) amine	TPA	TPA TMC
pentadentate	tetraethylene-pentamine	1,4,7,10,13-pentaazatridecane		$NH_2CH_2CH_2NHCH_2CH_2NHCH_2CH_2NHCH_2CH_2NH_2$
hexadentate	ethylenediamine-tetraacetate	1,2-ethanediyl (dinitrilo) tetraacetate	EDTA	

TABLE 4 Multidentate (Chelating) Ligands

Common Name	IUPAC Name	Abbreviation	Formula and Structure	
acetylacetonato	2,4-pentanedionato	acac	$CH_3COCHCOCH_3^-$	
2,2′-bipyridine	2,2′-bipyridyl	bipy	$C_{10}H_8N_2$	
nacnac	*N*,*N*′-diphenyl-2,4-pentanediiminato	nacnac	$C_{17}H_{17}N_2^-$	
1,10-phenanthroline, o-phenanthroline	1,10-diazaphenanthrene	phen, o-phen	$C_{12}H_8N_2$	
oxalato	oxalato	ox	$C_2O_4^{2-}$	
dialkyldithio-carbamato	dialkyl-carbamodithioato	dtc	$S_2CNR_2^-$	
ethylenedithiolate	1,2-ethenedithiolate	dithiolene	$S_2C_2H_2^{2-}$	
1,2-bis (diphenylphosphino) ethane	1,2-ethanediylbis-(diphenylphosphane)	dppe	$Ph_2PC_2H_4PPh_2$	
BINAP	2,2′-bis (diphenylphosphino) -1,1′-binaphthyl	BINAP	$Ph_2P(C_{10}H_6)_2PPh_2$	
dimethylglyoximato	butanedione dioxime	DMG	$HONCO(CH_3)C(CH_3)NO^-$	
pyrazolylborato (scorpionate)	hydrotris-(pyrazo-1-yl)borato	Tp	$[HB(C_3H_3N_2)_3]^-$	
salen	2,2′-Ethylenebis-(nitrilomethylidene)-diphenoxide	salen	$^-OPh(CHNCH_2CH_2NCH)PhO^-$	

3. The number of ligands of each kind is indicated by prefixes (in margin). In simple cases, the prefixes in the second column are used. If the ligand name already includes these prefixes or is complicated, it is set off in parentheses, and prefixes in the third column (ending in *–is*) are used.

Examples: dichlorobis(ethylenediamine)cobalt(III), $[Co(NH_2CH_2CH_2NH_2)_2Cl_2]^+$

tris(2,2′-bipyridine)iron(II), $[Fe(C_{10}H_8N_2)_3]^{2+}$

4. Ligands are generally written in alphabetical order—according to the ligand name, not the prefix.

Examples: tetraamminedichlorocobalt(III), $[Co(NH_3)_4Cl_2]^+$

(tetraammine is alphabetized by *a* and dichloro by *c*, not by the prefixes)

amminebromochloromethylamineplatinum(II), $Pt(NH_3)BrCl(CH_3NH_2)$

4. Where there are several ligands of the same kind, we normally use the prefixes di, tri, tetra, penta and hexa to show the number of ligands of that type. An exception occurs when the name of the ligand includes a number, e.g. dipyridyl or ethylenediamine. To avoid confusion in such cases, bis, tris and tetrakis are used instead of di, tri and tetra and the name of the ligand is placed in brackets.

2	di	bis
3	tri	tris
4	tetra	tetrakis
5	penta	pentakis
6	hexa	hexakis
7	hepta	heptakis
8	octa	octakis
9	nona	nonakis
10	deca	decakis

Examples: tris(tetraammine-μ-dihydroxocobalt)cobalt(6+),
$[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$
μ-amido-μ-hydroxobis(tetramminecobalt)(4+),
$[(NH_3)_4Co(OH)(NH_2)Co(NH_3)_4]^{4+}$

The oxidation state of the central metal is shown by a Roman numeral in brackets immediately following its name (i.e. no space, e.g. titanium(III)).

Complex positive ions and neutral molecules have no special ending but complex negative ions end in -ate.

5. Anionic ligands are given an *o* suffix. Neutral ligands retain their usual name. Coordinated water is called *aqua* and coordinated ammonia is called *ammine*. Examples are in Table 2.

6. Two systems exist for designating charge or oxidation number:
   a. The *Stock system* puts the calculated oxidation number of the metal as a Roman numeral in parentheses after the metal name. Although this is the most commonly employed method, its drawback is that the oxidation state of a metal within a complex can be ambiguous, and difficult to specify.
   b. The *Ewing-Bassett system* puts the charge on the coordination sphere in parentheses after the name of the metal. This convention offers an unambiguous identification of the species.

   In either case, if the charge is negative, the suffix *-ate* is added to the name.

Examples: tetraammineplatinum(II) or tetraammineplatinum(2+), $[Pt(NH_3)_4]^{2+}$

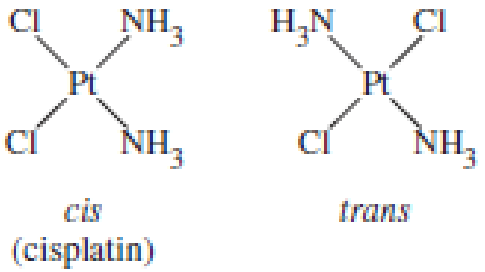
tetrachloroplatinate(II) or tetrachloroplatinate(2–), $[PtCl_4]^{2-}$

hexachloroplatinate(IV) or hexachloroplatinate(2–), $[PtCl_6]^{2-}$

7. Prefixes designate adjacent (*cis*-) and opposite (*trans*-) geometric locations (Figures 1 and 5). Other prefixes will be introduced as needed.

Examples: *cis*- and *trans*-diamminedichloroplatinum(II), $[PtCl_2(NH_3)_2]$

cis- and *trans*-tetraamminedichlorocobalt(III), $[CoCl_2(NH_3)_4]^+$



cis (cisplatin) *trans*

FIGURE 5 *cis* and *trans* Isomers of Diamminedichloroplatinum(II), $[PtCl_2(NH_3)_2]$. The *cis* isomer, also known as *cisplatin*, is used in cancer treatment.

If the complex contains two or more metal atoms, it is termed polynuclear_. The bridging ligands which link the two metal atoms together are indicated by the prefix μ-. If there are two or more bridging groups of the same kind, this is indicated by di-μ-, tri-μ- etc. Bridging groups are **listed alphabetically with the other groups unless the symmetry of the molecule allows a simpler name. If a bridging group bridges more than two metal atoms it is shown as μ3, μ4; μ5 or μ6 to indicate how many atoms it is bonded to.**

8. Sometimes a ligand may be attached through different atoms. Thus M-NO2 is called nitro and M-ONO is called nitrito. Similarly the SCN group may bond M-SCN thiocyanato or M-NCS isothiocyanato. These may be named systematically thiocyanato-S or thiocyanato-N to indicate which atom is bonded to the metal. This convention may be extended to other cases where the mode of linkage is ambiguous.

8. Bridging ligands between two metal ions (Figures 2 and 6) have the prefix μ-.

Examples: tris(tetraammine-μ-dihydroxocobalt)cobalt(6+), $[Co(Co(NH_3)_4(OH)_2)_3]^{6+}$

Shown Ewing –Basset system

μ-amido-μ-hydroxobis(tetramminecobalt)(4+), $[(NH_3)_4Co(OH)(NH_2)Co(NH_3)_4]^{4+}$

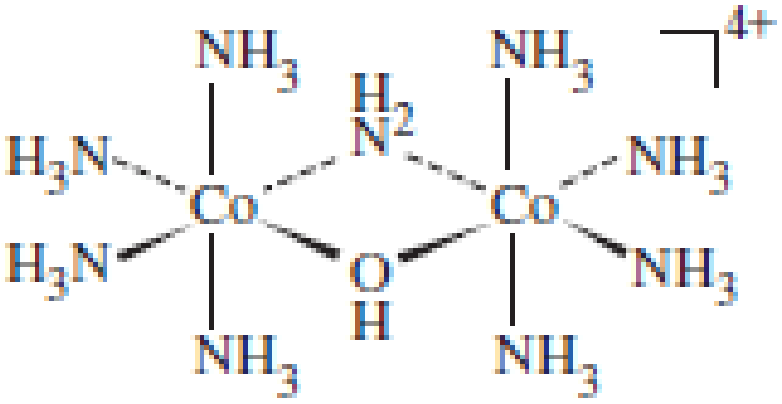


FIGURE 6 Bridging Amide and Hydroxide Ligands in μ-amido-μ-hydroxobis (tetraamminecobalt) (4+), $[(NH_3)_4Co(OH)(NH_2)Co(NH_3)_4]^{4+}$.

Shown Ewing –Basset system

9. If any lattice components such as water or solvent of crystallization are present, these follow the name, and are preceded by the number Of these groups in Arabic numerals.

9. When the complex is negatively charged, the names for these metals are derived from the sources of their symbols:

iron (Fe)	ferrate	lead (Pb)	plumbate
silver (Ag)	argentate	tin(Sn)	stannate
		gold (Au)	aurate

Examples: tetrachloroferrate(III) or tetrachloroferrate(1–), $[FeCl_4]^-$

dicyanoaurate(I) or dicyanoaurate(1–), $[Au(CN)_2]^-$

Complex ~~anions~~ cation	
$[Co(NH_3)_6]Cl_3$	Hexaamminecobalt(III) chloride
$[CoCl(NH_3)_5]^{2+}$	Pentaamminechlorocobalt(III) ion
$[CoSO_4(NH_3)_4]NO_3$	Tetraamminesulphatocobalt(III) nitrate
$[Co(NO_2)_3(NH_3)_3]$	Triamminetrinitrocobalt(III)
$[CoCl \cdot CN \cdot NO_2 \cdot (NH_3)_3]$	Triamminechlorocyanonitrocobalt(III)
$[Zn(NCS)_4]^{2+}$	Tetrathiocyanato-N-zinc(II)
$[Cd(SCN)_4]^{2+}$	Tetrathiocyanato-S-cadmium(II)
Complex ~~cations~~ Anion.	
$Li[AlH_4]$	Lithium tetrahydridoaluminate(III) (*lithium aluminium hydride*)
$Na_2[ZnCl_4]$	Sodium tetrachlorozincate(II)
$K_4[Fe(CN)_6]$	Potassium hexacyanoferrate(II)
$K_3[Fe(CN)_5NO]$	Potassium pentacyanonitrosylferrate(II)
$K_2[OsCl_5N]$	Potassium pentachloronitridoosmate(VI)
$Na_3[Ag(S_2O_3)_2]$	Sodium bis(thiosulphato)argentate(I)
$K_2[Cr(CN)_2O_2(O_2)NH_3]$	Potassium amminedicyanodioxoperoxo chromate(VI)

Organic groups

$[Pt(py)_4][PtCl_4]$	Tetrapyridineplatinum(II) · tetrachloroplatinate(II)
$[Cr(en)_3]Cl_3$	*d* or *l* Tris(ethylenediamine)chromium(III) chloride
$[CuCl_2(CH_3NH_2)_2]$	Dichlorobis(dimethylamine)copper(II)
$Fe(C_5H_5)_2$	Bis(cyclopentadienyl)iron(II)
$[Cr(C_6H_6)_2]$	Bis(benzene)chromium(0)

Bridging groups	
$[(NH_3)_5Co \cdot NH_2 \cdot Co(NH_3)_5](NO_3)_5$	μ-amidobis[pentaamminecobalt(III)] nitrate
$[(CO)_3Fe(CO)_3Fe(CO)_3]$	Tri-μ-carbonyl-bis(tricarbonyliron(0)) (*di iron enneacarbonyl*)
$[Be_4O(CH_3COO)_6]$	Hexa-μ-acetato(O,O′)-μ_4-oxo-tetraberyllium(II) (*basic beryllium acetate*)
Hydrates	
$AlK(SO_4)_2 \cdot 12H_2O$	Aluminium potassium sulphate 12-water

WERNER'S WORK

Werner's coordination theory in 1893 was the first attempt to explain the bonding in coordination complexes. It must be remembered that this imaginative theory was put forward before the electron had been discovered by J.J. Thompson in 1896, and before the electronic theory of valency. This theory and his painstaking work over the next 20 years won Alfred Werner the Nobel Prize for Chemistry in 1913.

Complexes must have been a complete mystery without any knowledge of bonding or structure. For example, why does a stable salt like $CoCl_3$ react with a varying number of stable molecules of a compound such as NH_3 to give several new compounds: $CoCl_3 \cdot 6NH_3$, $CoCl_3 \cdot 5NH_3$ and $CoCl_3 \cdot 4NH_3$? What are their structures? At that time X-ray diffraction, which is the most powerful method of determining the structures of crystals, had yet to be discovered. Werner did not have at his disposal any of the modern instrumental techniques, and all his studies were made using simple reaction chemistry. *Werner was able to explain the nature of bonding in complexes, and he concluded that in complexes the metal shows two different sorts of valency:*

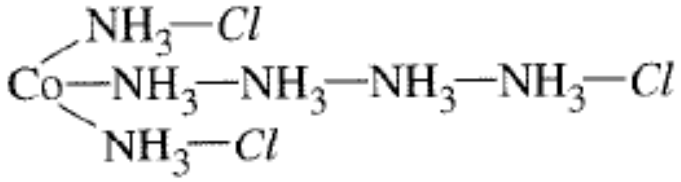
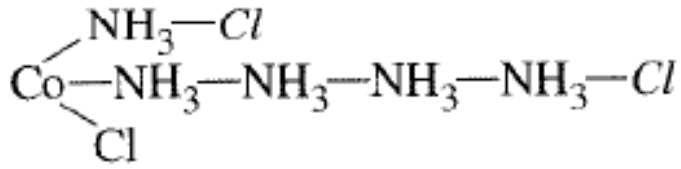
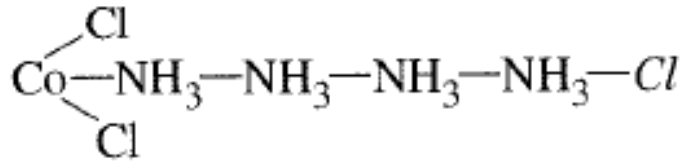
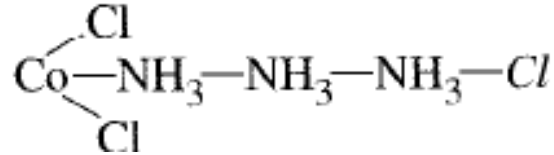
Early in the twentieth century, Werner and Jørgensen made considerable progress on understanding the coordination chemistry of transition metals and also discovered a number of organometallic compounds. Nevertheless, the popularity of inorganic chemistry as a field of study gradually declined during most of the first half of the century.

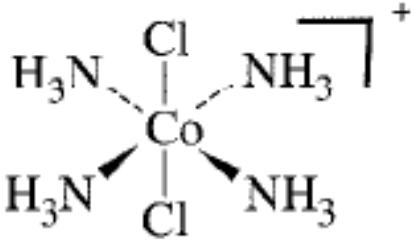
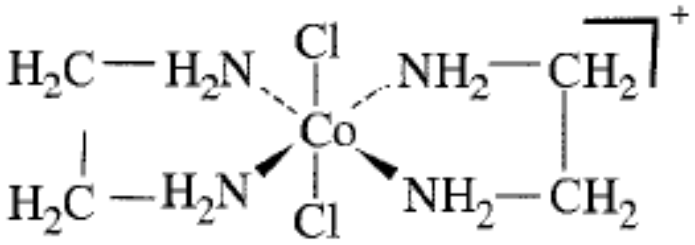
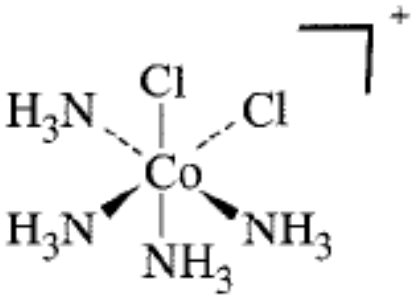
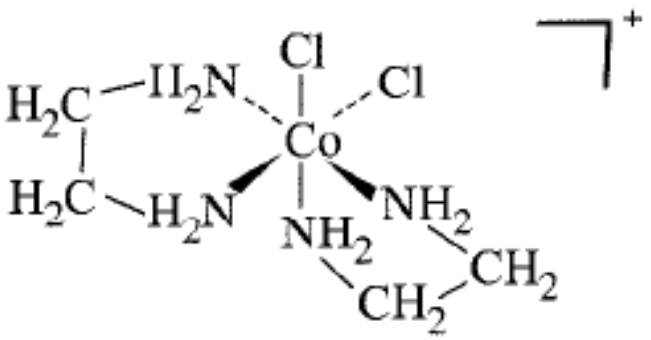
The need for inorganic chemists to work on military projects during World War II rejuvenated interest in the field. As work was done on many projects (not least of which was the Manhattan Project, in which scientists developed the fission bomb), new areas of research appeared, and new theories were proposed that prompted further experimental work.

A great expansion of inorganic chemistry began in the 1940s, sparked by the enthusiasm and ideas generated during World War II.

Inorganic chemists tried to use the advances in organic bonding theory and the simple ideas of ionic charges to explain bonding in coordination compounds, but found that the theories were inadequate. In a compound such as hexaamminecobalt(III) chloride, $[Co(NH_3)_6]Cl_3$, the early bonding theories allowed only three other atoms to be attached to the cobalt (because of its "valence" of 3). By analogy with ordinary salts, such as $FeCl_3$, the chlorides were assigned this role. This left the six ammonia molecules with no means of participating in bonding, and it was necessary to develop new ideas to explain the structure. One theory, proposed first by C. W. Blomstrand[1] (1826–1894) and developed further by S. M. Jørgensen[2] (1837–1914), was that the nitrogens could form chains much like those of carbon (and thus could have a valence of 5) as shown in Table 9-1, and that chloride ions attached directly to cobalt were bonded more strongly than those bonded to nitrogen. Alfred Werner[3] (1866–1919) proposed instead that all six ammonias could bond directly to the cobalt ion. Werner allowed for a looser bonding of the chloride ions; we now consider them as independent ions. The series of compounds in Table 9-1 illustrates how both the chain theory and Werner's coordination theory predict the number of ions to be formed by a series of cobalt complexes. Blomstrand's theory allowed dissociation of chlorides attached to ammonia but not of chlorides attached directly to cobalt. Werner's theory also included two kinds of chlorides. The number of chlorides attached to the cobalt (and therefore unavailable as ions) plus the number of ammonia molecules totaled six. The other chlorides were considered less firmly bound and could therefore form ions in solution. We now consider them to be ions in the solid state as well.

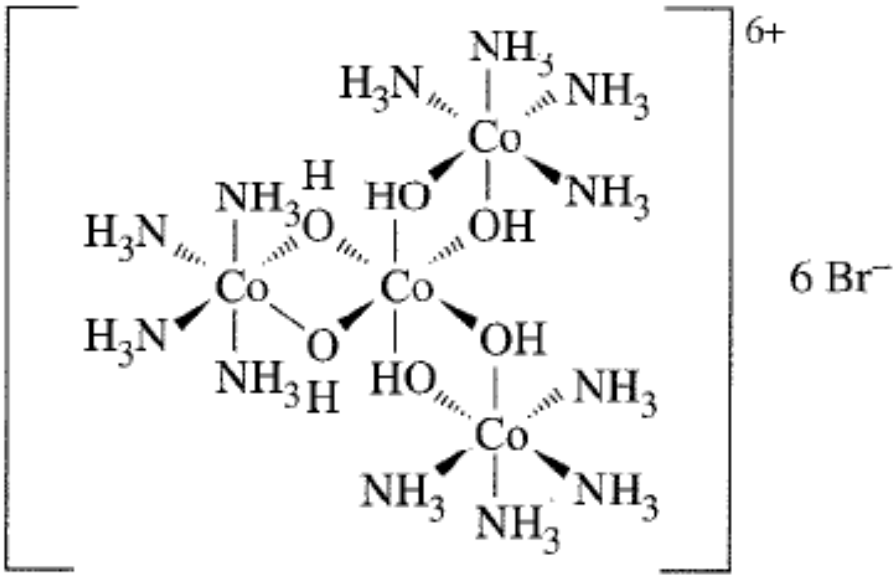
TABLE 9-1
Comparison of Blomstrand's Chain Theory and Werner's Coordination Theory

Werner Formula (Modern Form)	*Number of Ions Predicted*	*Blomstrand Chain Formula*	*Number of Ions Predicted*
$[Co(NH_3)_6]Cl_3$	4		4
$[Co(NH_3)_5Cl]Cl_2$	3		3
$[Co(NH_3)_4Cl_2]Cl$	2		2
$[Co(NH_3)_3Cl_3]$	0		2

	$[Co(NH_3)_4Cl_2]^+$	$[Co(H_2NC_2H_4NH_2)_2Cl_2]^+$
trans green		
cis violet		

Except for the last compound in the table, the predictions match, and the ionic behavior does not distinguish between them. Even with the last compound, problems with purity and conductance measurements left some ambiguity. The argument between Jørgensen and Werner continued for many years, with each presenting data and explanations favoring his own position. This case illustrates some of the good features of such controversy. Werner was forced to develop his theory further and synthesize new compounds to test his ideas because Jørgensen defended the earlier theory so vigorously. Werner proposed an octahedral structure for compounds such as those in Table 9-1. He prepared and characterized many isomers, including both green and violet forms of $[Co(H_2NC_2H_4NH_2)_2Cl_2]^+$. He claimed that these compounds had the chlorides arranged ***trans*** (opposite each other) and ***cis*** (adjacent to each other) respectively, in an overall octahedral geometry, as in Figure 9-1. Jørgensen offered alternative isomeric structures but finally conceded defeat in 1907, when Werner succeeded in synthesizing the green *trans* and the violet *cis* isomers of $[Co(NH_3)_4Cl_2]^+$, for which there were no counterparts in the chain theory.

However, even synthesis of this compound and the later discovery of optically active coordination compounds did not completely convince all chemists, although such compounds could not be explained directly by the chain theory. It was argued that Werner's optically active compounds still contained carbon, and that their chirality could be due to the carbon atoms. Finally, Werner resolved the compound $[Co(Co(NH_3)_4(OH)_2)_3]Br_6$ (Figure 9-2), initially prepared by Jørgensen, into its two optically active forms, using *d*- and *l*-α-bromocamphor-π-sulfonate as the resolving agents. With this final proof of optical activity without carbon, the validity of Werner's theory was finally accepted. Pauling[4] extended the theory in terms of hybrid orbitals, and later theories[5] have adapted arguments first used for electronic structures of ions in crystals to coordination compounds.



Optical activity in coordination complex
First time ever seen

Postulates of Werner's Theory

The important postulates of Werner's theory are:

- The central metal or the metal atoms in coordination compounds show two types of valency. They are the primary and secondary valency.
- The primary valency relates to the oxidation state and the secondary valency relates to the coordinate number.
- The number of secondary valences is fixed for every metal atom. It means that the coordination number is fixed.
- The metal atom works towards satisfying both its primary and secondary valencies. A negative ion satisfies the primary valency. On the other hand, a negative ion or neutral molecules satisfy secondary valencies.
- The secondary valencies point towards a fixed position in space. This is the reason behind the definite geometry of the coordinate compound. For example, let us consider the case of a metal ion having six secondary valencies. These arrange octahedrally around the central metal ion. If the metal ion has four secondary valencies, these arrange in either tetrahedral or square planar arrangement around the central metal ion. Therefore, we see that the secondary valency determines the stereochemistry of the complex ion. On the other hand, the primary valency is non-directional.

Q: What are the limitations of Werner's Theory?

Ans: Like all the major theories, Werner's Theory was not free from limitations. The common limitations of the theory are:

- It could not explain the inability of all elements to form coordination compounds.
- The Werners theory could not explain the directional properties of bonds in various coordination compounds.
- It does not explain the colour, the magnetic and optical properties shown by coordination compounds.

of electrons from nitrogen into the empty orbitals of boron. The coordinate covalent bond is shown by an arrow with its head pointing towards the direction of the donation of an electron pair, as shown in Figure 2.

A **complex** is a molecule/ion containing a central metal atom/ion surrounded by a definite number of ligands held by secondary valences or coordinate covalent bonds.

Primary valency refers to the charge over the metal ion e.g. Co(III) has +3 charge, which can be balanced by −3 charge-forming compounds like $CoCl_3$. The primary valency is ionic and is satisfied in the second coordination sphere, as shown in Figure 3.

Secondary valency is the number of empty valence orbitals, as illustrated for $[Co(NH_3)_6]Cl_3$ in the figure. The Co(III) ion has six empty valence orbitals. Hence its secondary valency is six. Secondary valency is a coordinate covalent valency, and it is satisfied in the first coordination sphere of the metal ion, as shown in Figure 4.

Coordination number is a property of the metal ion representing the total number of donor atoms directly attached to the central atom. In the above case, the coordination number of Co(III) is six, as six nitrogen donor atoms are directly connected to the central metal ion (cobalt(III)).

Ligand is any atom, ion or neutral molecule capable of donating an electron pair and bonded to the central metal ion or atom through secondary valency.

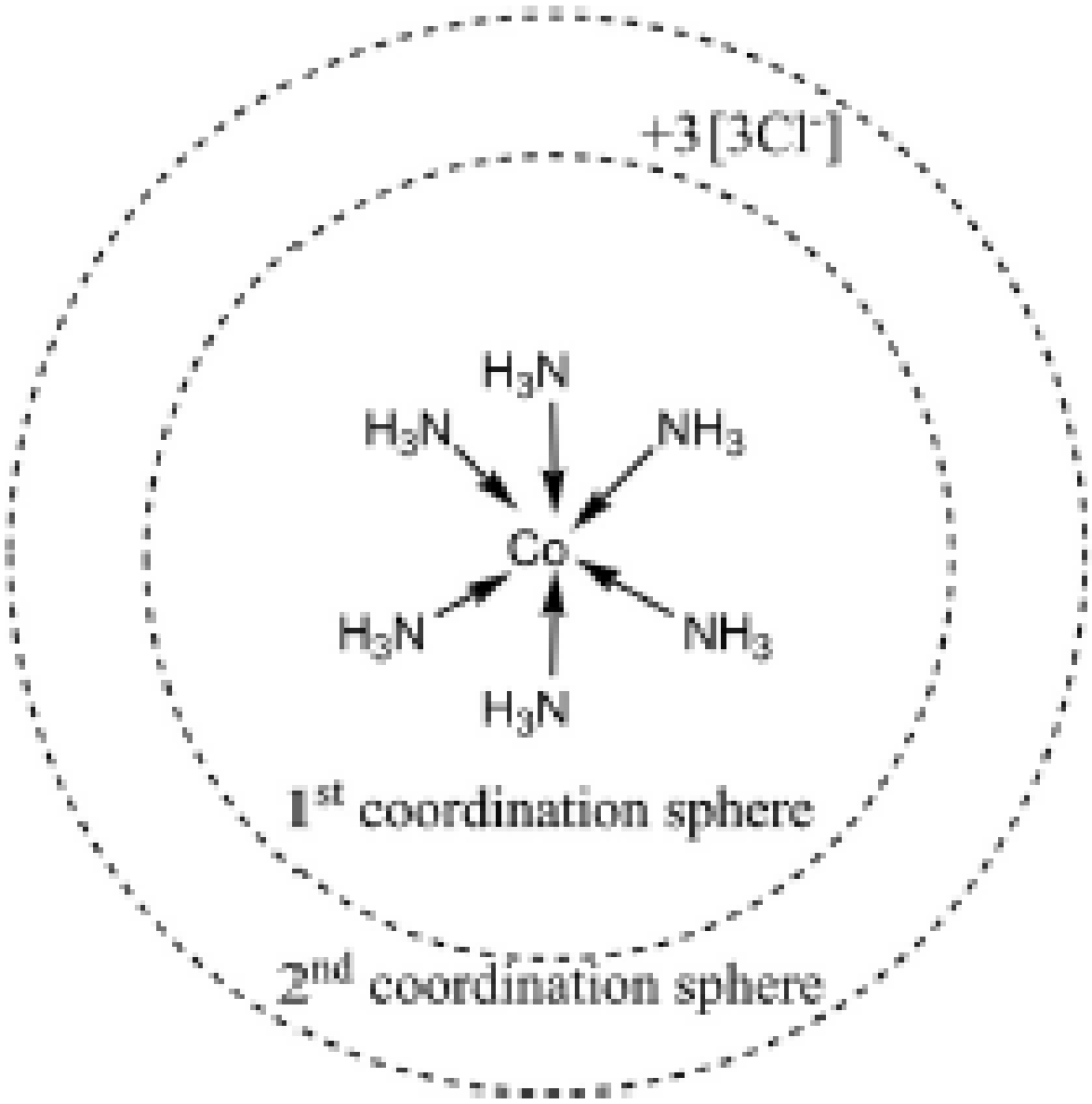


Figure 3 First and second coordination spheres in $[Co(NH_3)_6]Cl_3$.

This article is about the co-ordination of a ligand to a central atom. For the sense of touch in technology, see Haptic technology.

Hapticity is the coordination of a ligand to a metal center via an uninterrupted and contiguous series of atoms.[1] The hapticity of a ligand is described with the Greek letter η ('eta'). For example, η^2 describes a ligand that coordinates through 2 contiguous atoms. In general the η-notation only applies when multiple atoms are coordinated (otherwise the κ-notation is used). In addition, if the ligand coordinates through multiple atoms that are **not** contiguous then this is considered denticity[2] (not hapticity), and the κ-notation is used once again.[3] When naming complexes care should be taken not to confuse η with μ ('mu'), which relates to bridging ligands.[4][5]

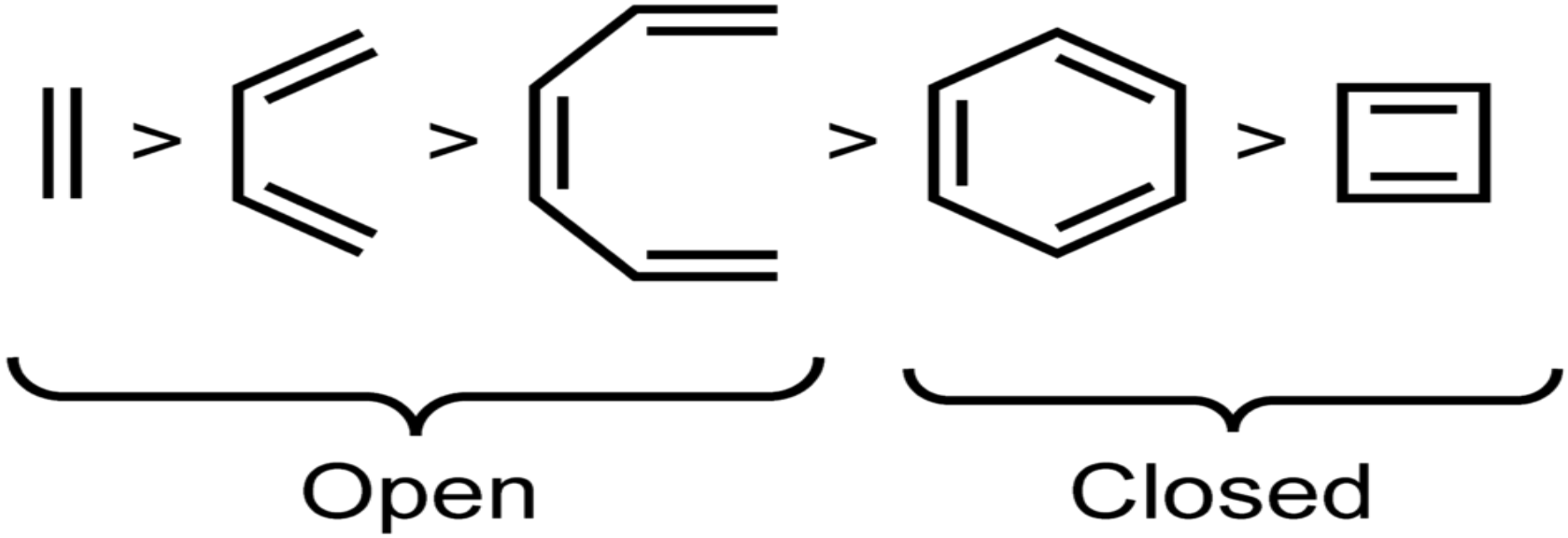
Side-on bonded ligands containing multiple bonded atoms, e.g. ethylene in Zeise's salt or with fullerene, which is bonded through donation of the π-bonding electrons: $K[PtCl_3(\eta^2\text{-}C_2H_4)]\cdot H_2O$

Hapticity vs Denticity

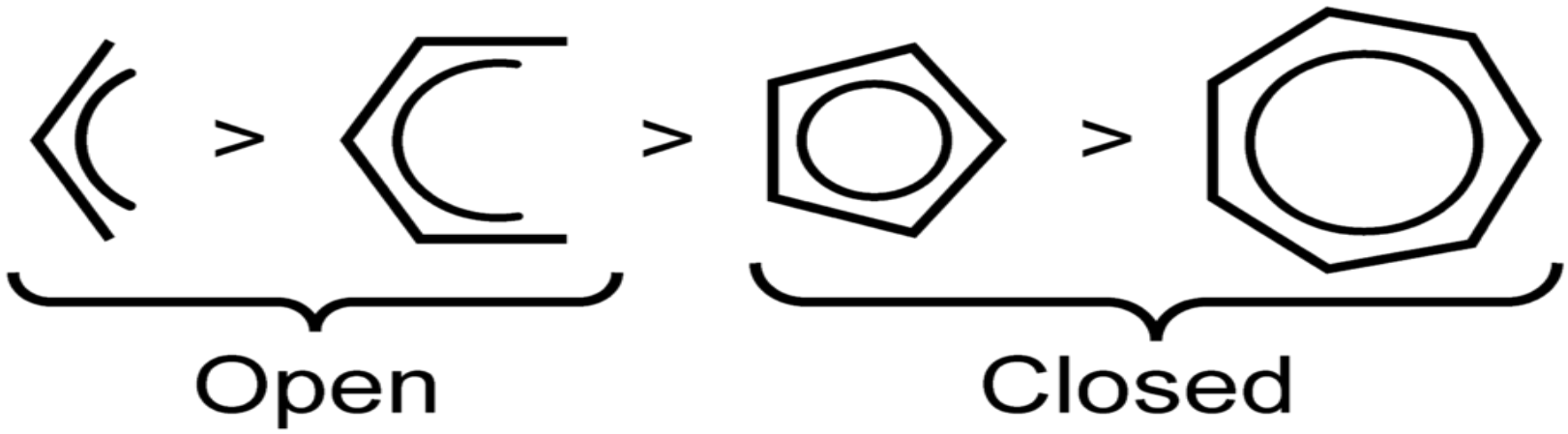
More Information Online WWW.DIFFERENCEBETWEEN.COM

	Hapticity	Denticity
DEFINITION	Hapticity is the coordination of a ligand to a metal center via uninterrupted and contiguous series of atoms	Denticity is the number of donor groups of the same ligand that binds with a metal center
THEORY	Gives the number of contiguous atoms involved in the coordination process	Gives the number of donor groups of the ligand attached with a metal center
NOTATION	η-notation	k-notation
EXAMPLES	Ferrocene has two ligands bound to Fe center and each ligand binds through 5 contiguous atoms	EDTA coordinates with metal centers via 6 non-contiguous atoms

Even hapticity



Odd hapticity

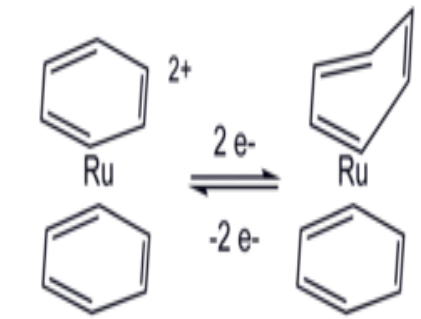


Electrons donated by "π-ligands" versus hapticity [edit]

Ligand	Electrons contributed (neutral counting)	Electrons contributed (ionic counting)
η^1-allyl	1	2
η^3-allyl cyclopropenyl	3	4
η^2-butadiene	2	2
η^4-butadiene	4	4
η^1-cyclopentadienyl	1	2
η^3-cyclopentadienyl	3	4
η^5-cyclopentadienyl pentadienyl cyclohexadienyl	5	6
η^2-benzene	2	2
η^4-benzene	4	4
η^6-benzene	6	6
η^7-cycloheptatrienyl	7	6 or 10
η^8-cyclooctatetraenyl	8	10

Changes in hapticity [edit]

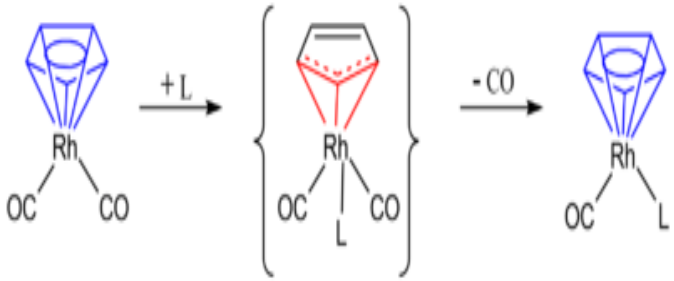
The hapticity of a ligand can change in the course of a reaction.[12] E.g. in a redox reaction:



$[Ru(\eta^6\text{-}C_6H_6)_2]^{2+}$ $Ru(\eta^6\text{-}C_6H_6)(\eta^4\text{-}C_6H_6)$

Here one of the η^6-benzene rings changes to a η^4-benzene.

Similarly hapticity can change during a substitution reaction:



Here the η^5-cyclopentadienyl changes to an η^3-cyclopentadienyl, giving room on the metal for an extra 2-electron donating ligand 'L'. Removal of one molecule of CO and again donation of two more electrons by the cyclopentadienyl ligand restores the η^5-cyclopentadienyl. The so-called indenyl effect also describes changes in hapticity in a substitution reaction.

κ (kappa)

in inorganic nomenclature

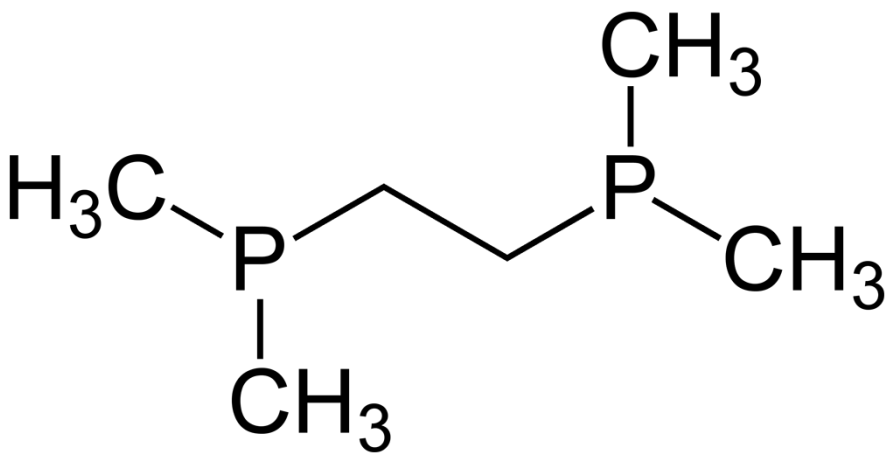
Online use...

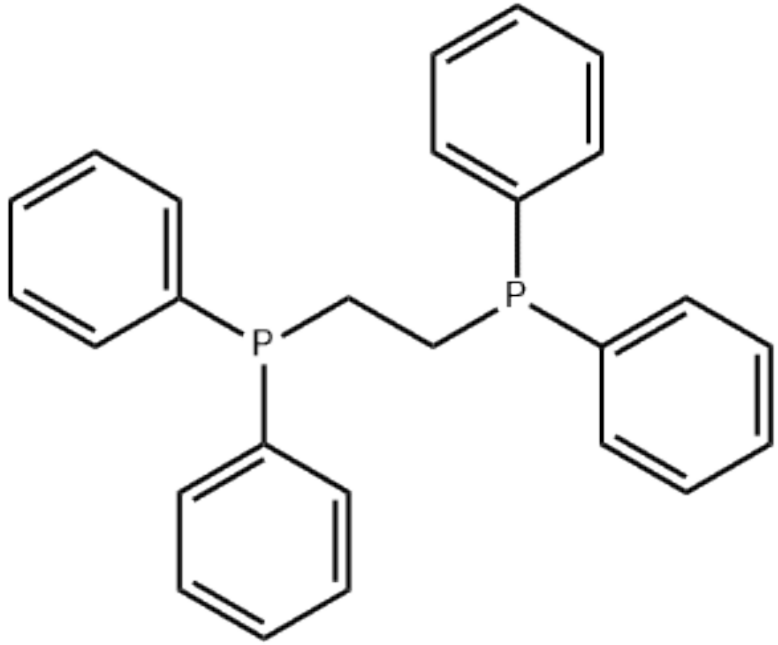
https://doi.org/10.1351/goldbook.K03366

An affix in the name of a polydentate chelate complex indicating single ligating atom attachments of a polyatomic ligand to a coordination centre. A right superscript numerical index indicates the number of such attachments. Example: $Ni[(CH_3)_2PCH_2CH_2P(CH_3)_2]Br_2$, dibromobis[1,2-ethanediylbis(dimethylphosphine)-κ^2P]nickel(II).

Source:

Red Book, 3rd ed., p. 174 [Terms] [Book]





3. THEORIES OF BONDING IN COORDINATION COMPOUNDS

The theories of bonding in coordination compounds [3] have evolved subsequent to Werner's coordination theory (1893). Werner introduced the concept of primary and secondary valency, explaining the formation of the coordination compounds. The 18-electron rule, stating that the stable complexes with low formal oxidation states of metal ions should have 18 bonding electrons around the metal ion, became an important beginning point toward the study of the stability of the complexes. The 18-electron rule is significant in modern coordination chemistry as it is also supported by the molecular orbital theory. However, a smaller number of complexes with metals in low oxidation states restrict its wide applicability. An important advance in the theories of bonding in coordination compounds was the introduction of

valence bond theory (VBT), which is actually an extension of Lewis theory (1902) and Heitler–London theory (1927). From the above two theories, the sharing of electron pairs by atoms and the formation of bonds was understood and explained. The VBT developed by Linus Pauling introduced two key concepts in the theories of bonding, viz, the hybridization of orbitals and resonance, the absence of which were the drawbacks of the previous theories.

According to this theory, the metal ions are regarded as Lewis acids characterized by the availability of low lying, empty orbitals suitable for accommodating the accepted electrons. The ligands are the Lewis bases, characterized by the availability of a lone pair of electrons that can be readily donated, resulting into a formation of a coordinate covalent bond. Thus, the acid–base characteristics of metal and ligand complement each other to give rise to a coordinate covalent bond, as shown in Figure 8.

Hybridization of atomic orbitals, a mathematical tool for mixing the atomic orbitals to give an equal number of hybrid orbitals with superior directional properties, became a very important feature of this theory in explaining various geometries observed in coordination compounds.

1.1 Definitions

Coordination compounds are the compounds containing one or more coordinate covalent bonds.

Coordinate covalent bonds are the covalent bonds in which both the bonding electrons are contributed by one of the bond partners. Figure 2 distinguishes the covalent bonds from the coordinate covalent bond in NH_3BF_3. While the three B–F covalent bonds are formed due to the sharing of electron pairs resulting from contributions of both boron and fluorine atoms, an N–B bond is formed due to the donation of a lone pair

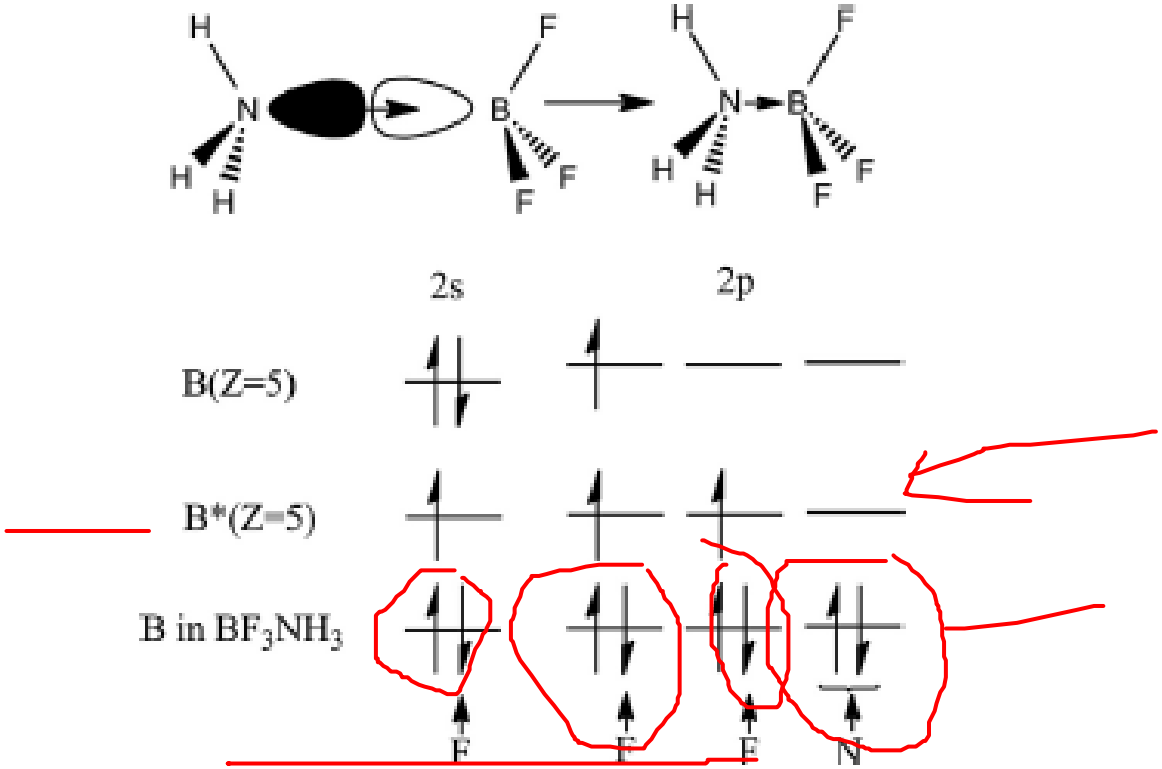


Figure 2 Bonding in NH_3BF_3.

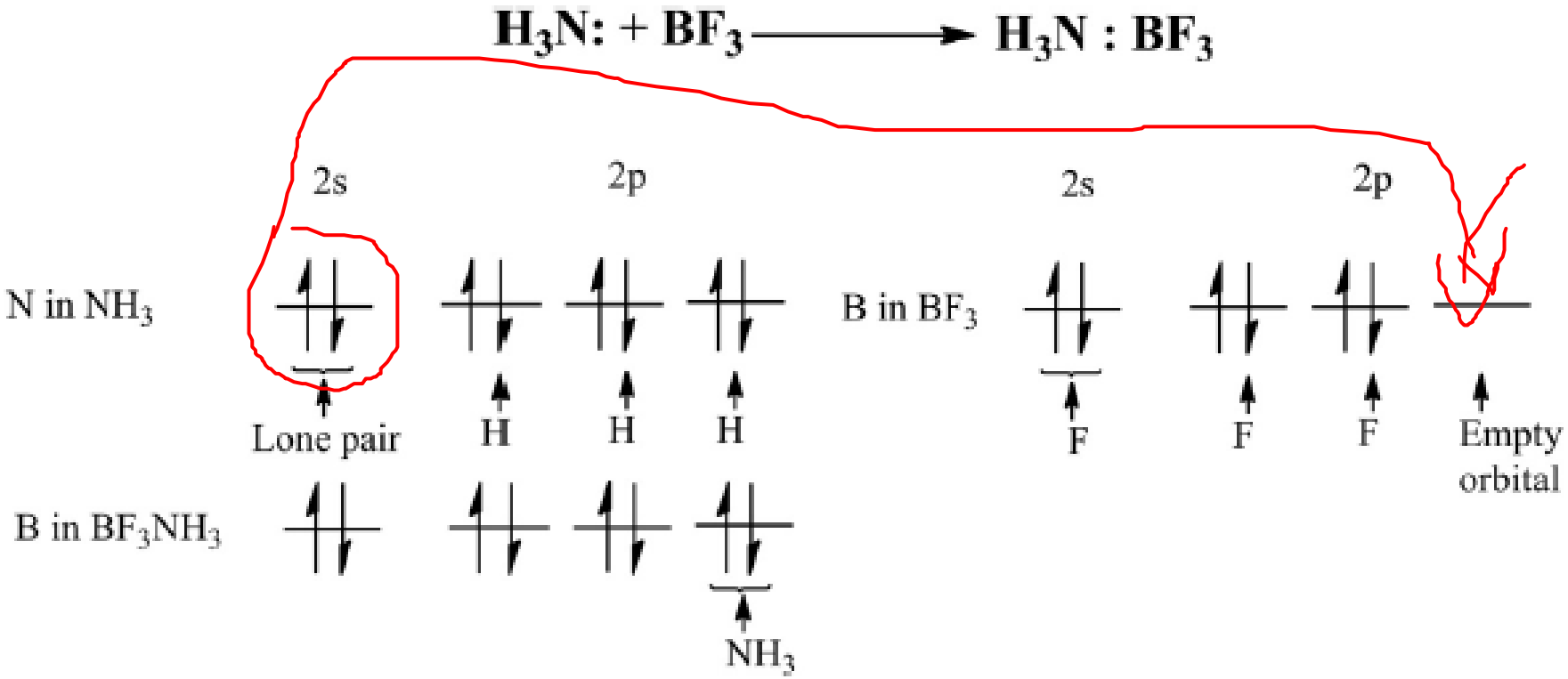


Figure 8 Covalent and coordinate covalent bonding in BF_3NH_3.

Dentate character is a property of a ligand representing a number of coordinating atoms.

In the case of $[Co(NH_3)_6]Cl_3$, ammonia, NH_3 the ligand contains one donor atom (N). Hence its dentate character is one and is classified as a monodentate ligand. Similarly, chloro (Cl^-) is an anionic, monoatomic and monodentate ligand, while hydroxo (OH^-) is a diatomic, monodentate and anionic ligand. Aquo (OH_2) represents a neutral triatomic monodentate ligand. A few popular ligands and their characteristics are shown in Figure 5.

Due to a higher dentate character of ligands, a variety of complexes known as chelate is also formed sometimes. **Chelate** is a compound formed when a polydentate ligand uses more than one of its coordinating atoms to form a closed-ring structure, which includes the central metal ion. Five- and six-membered rings are known to provide extra stability to the chelates. The process of chelate formation is known as chelation. A polydentate ligand involved in chelate formation is also known as a chelating ligand. Chelates generally exhibit higher stability than analogous complexes.

A polydentate ligand may be attached to the central metal ion through more than one kind of functional group. The number and kind of linkages by which the metal ion is attached with the ligands can thus become a criterion for the classification of chelates. The covalent bonds are formed by the replacement of one or more H-atoms, while coordinate covalent bonds are formed by the donation of an electron pair from the ligands. Some of the chelates involving a variety of polydentate ligands and linkages are shown in Figure 6. The coordinate covalent linkages are shown by thin, thread-like bonds.

Sidgwick's electronic theory:



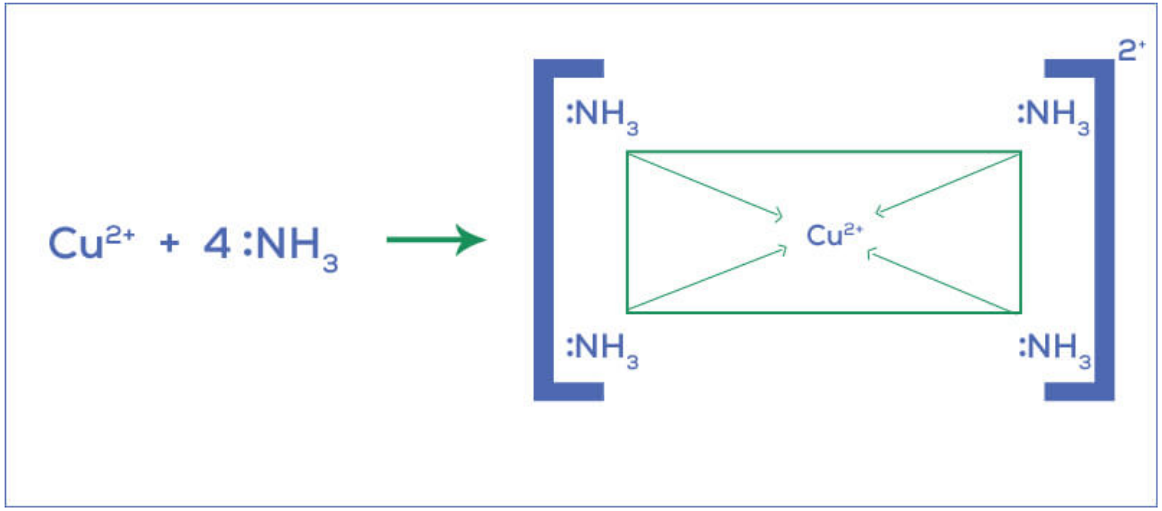
Nevil Sidgwick

(1873 – 1952)

In 1927, Sidgwick extended the electronic theory of lewis for the formation of coordination compounds. He introduced a new concept of a 'coordinate bond'.

According to Sidgwick's concept, the ligands donate the electron pair to the central metal ion and thus form a number of coordinate bonds.

Example: Cuprammonium ion is formed by the donation of four electron pairs by four ammonia molecules towards cupric ion.



Merits of Sidgwick theory:

- It explains the structure of complexes on the basis of electronic theory.
- It provides the idea of EAN (Effective atomic number) rule which can be used to find the number of ligands around a metal ion in a complex.

Demerits of Sidgwick theory:

- It accounts for the unnecessary accumulation of negative charges on the metal ion. This fact can not be accepted because metal ion cannot have a negative charge.
- The 2s orbitals present in the donating atoms of the ligands have no bonding property with the d-orbitals of the central metal ion.

FULL ARTICLE

Related Topics: coordination compound • Electron • Metal • Effective atomic number rule

Effective atomic number (EAN), number that represents the total number of electrons surrounding the nucleus of a metal atom in a metal complex. It is composed of the metal atom's electrons and the bonding electrons from the surrounding electron-donating atoms and molecules. Thus, the effective atomic number of the cobalt atom in the complex $[Co(NH_3)_6]^{3+}$ is 36, the sum of the number of electrons in the trivalent cobalt ion (24) and the number of bonding electrons from six surrounding ammonia molecules, each of which contributes an electron pair ($2 \times 6 = 12$).

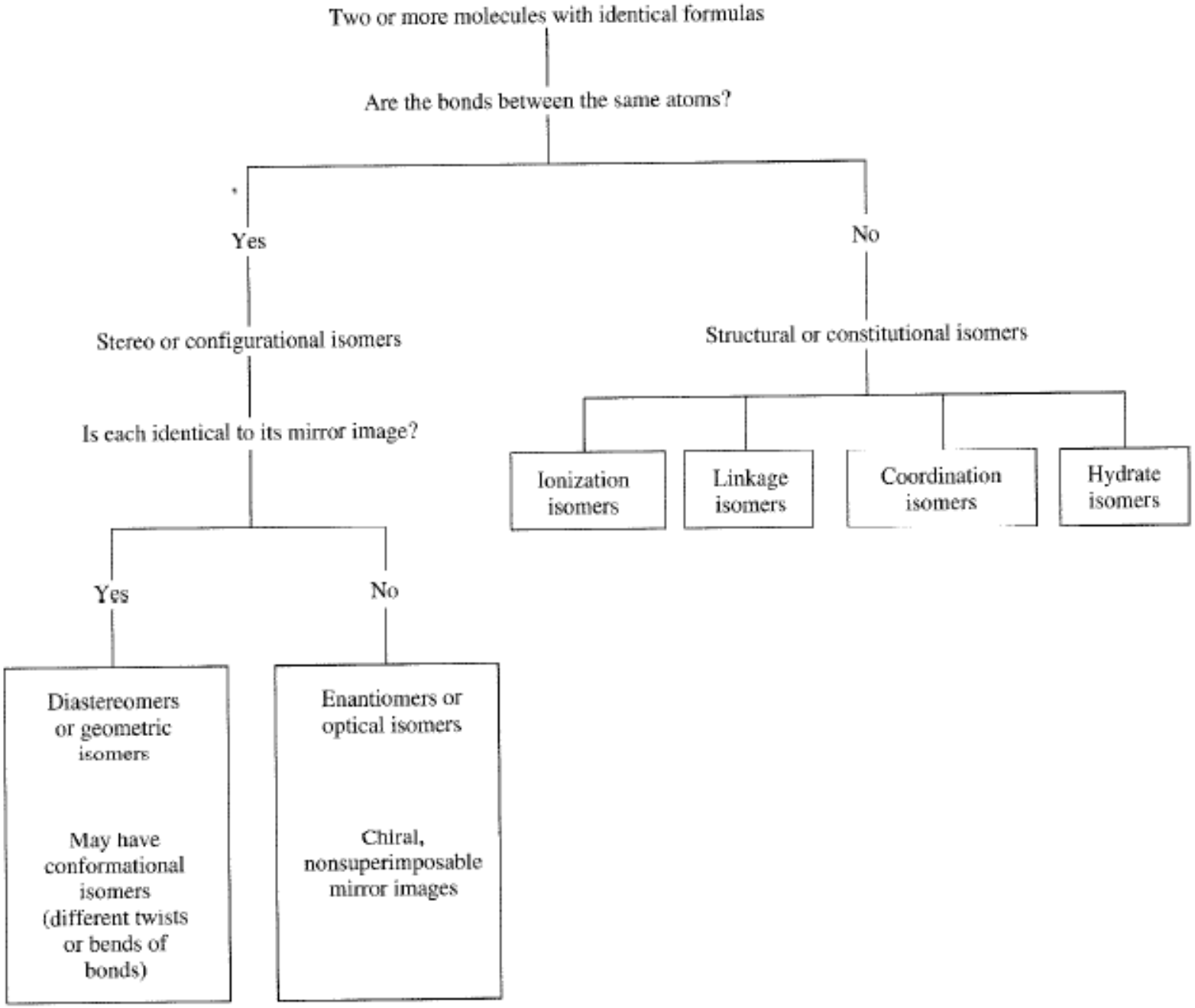
The English chemist Nevil V. Sidgwick made the observation, since known as the EAN rule, that in a number of metal complexes the metal

effective atomic number is numerically equal to the atomic number of the noble-gas element found in the same period in which the metal is situated. This rule seems to hold for most of the metal complexes with carbon monoxide, the metal carbonyls, as well as many organometallic compounds. By using this rule, it is possible to predict the number of ligands in these types of compounds and also the products of their reactions. The EAN rule is often referred to as the "18-electron rule" since, if one counts only valence electrons (6 for Co^{3+} and $2 \times 6 = 12$ for 6 NH_3), the total number is 18.

9-3-1 STEREOISOMERS

Stereoisomers include *cis* and *trans* isomers, chiral isomers, compounds with different conformations of chelate rings, and other isomers that differ only in the geometry of attachment to the metal ion. As mentioned at the beginning of this chapter, study of stereoisomers provided much of the experimental evidence used by Werner to develop and defend his coordination theory. Similar study of new compounds is useful in establishing structures and reactions, even though development of experimental methods such as automated X-ray diffraction can shorten the process considerably.

Isomers in coordination chemistry include many types. **Hydrate** or **solvent isomers**, **ionization isomers**, and **coordination isomers** have the same overall formula but have different ligands attached to the central atom or ion. The names indicate whether solvent, anions, or other coordination compounds form the changeable part of the structure. The terms **linkage isomerism** or **ambidentate isomerism** are used for cases of bonding through different atoms of the same ligand. **Stereoisomers** have the same ligands, but differ in the geometric arrangement of the ligands. The diagram and examples that follow may help make the distinctions clearer.


Two or more molecules with identical formulas
Are the bonds between the same atoms?
Yes
No
Stereo or configurational isomers
Structural or constitutional isomers
Is each identical to its mirror image?
Ionization isomers
Linkage isomers
Coordination isomers
Hydrate isomers
Yes
No
Diastereomers or geometric isomers
May have conformational isomers (different twists or bends of bonds)
Enantiomers or optical isomers
Chiral, nonsuperimposable mirror images

9-3-2 FOUR-COORDINATE COMPLEXES

Square-planar complexes may have *cis* and *trans* isomers as shown in Figure 9-4, but no chiral isomers are possible when the molecule has a mirror plane (as do many square-planar molecules). In making decisions about whether a molecule has a mirror plane, we usually ignore minor changes in the ligand such as rotation of substituent groups, conformational changes in ligand rings, and bending of bonds. Examples of square-planar complexes that do have chiral isomers are (*meso*-stilbenediamine)(*iso*-butylenediamine)platinum(II) and palladium(II) (Figure 9-7). In this case, the geometry of the

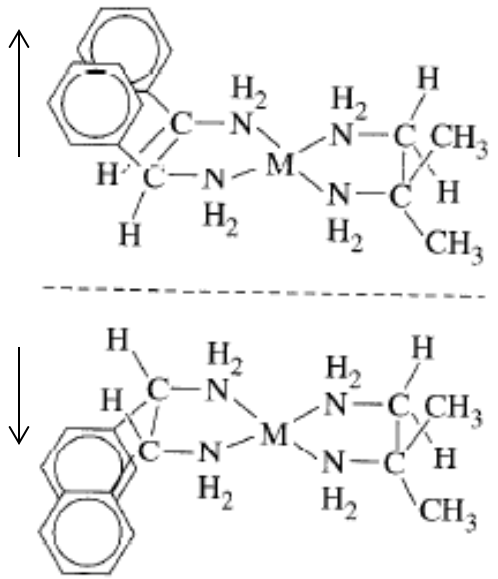
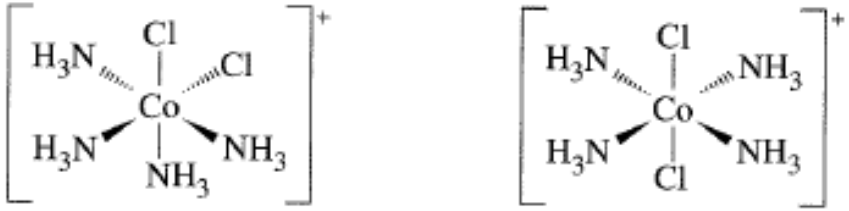


FIGURE 9-7 Chiral Isomers of Square-Planar Complexes: (*meso*-stilbenediamine)(*iso*-butylenediamine)platinum(II) and palladium(II). (From W. H. Mills and T. H. H. Quibell, *J. Chem. Soc.*, **1935**, 839; A. G. Lidstone and W. H. Mills, *J. Chem. Soc.*, **1939**, 1754.)

Trigonal antiprismatic (three isomers)

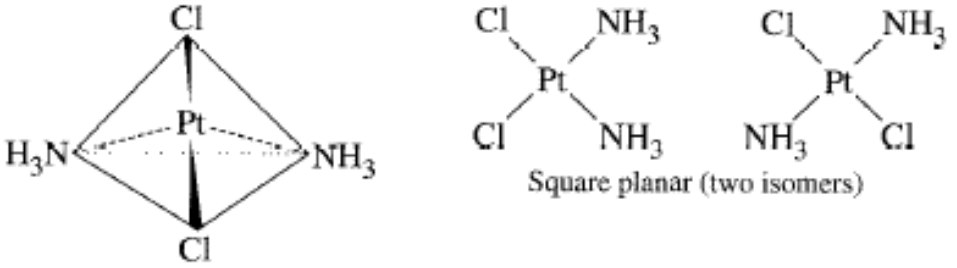


Octahedral (two isomers)

6-coordinate

FIGURE 9-3 Possible Isomers for Hexacoordinate Complexes.

cis - and *trans* - Diamminedichloroplatinum (II), $[PtCl_2(NH_3)_2]$



Tetrahedral (one isomer)

Square planar (two isomers)

4-coordinate

FIGURE 9-4 Possible Structures for Tetracoordinate Complexes.

9-3-4 SIX-COORDINATE COMPLEXES

Complexes of the formula $ML_3L'_3$, where L and L′ are monodentate ligands, may have two isomeric forms called *fac*- and *mer*- (for facial and meridional). *Fac* isomers have three identical ligands on one triangular face; *mer* isomers have three identical ligands in a plane bisecting the molecule. Similar isomers are possible with some chelating ligands. Examples with monodentate and tridentate ligands are shown in Figure 9-8.

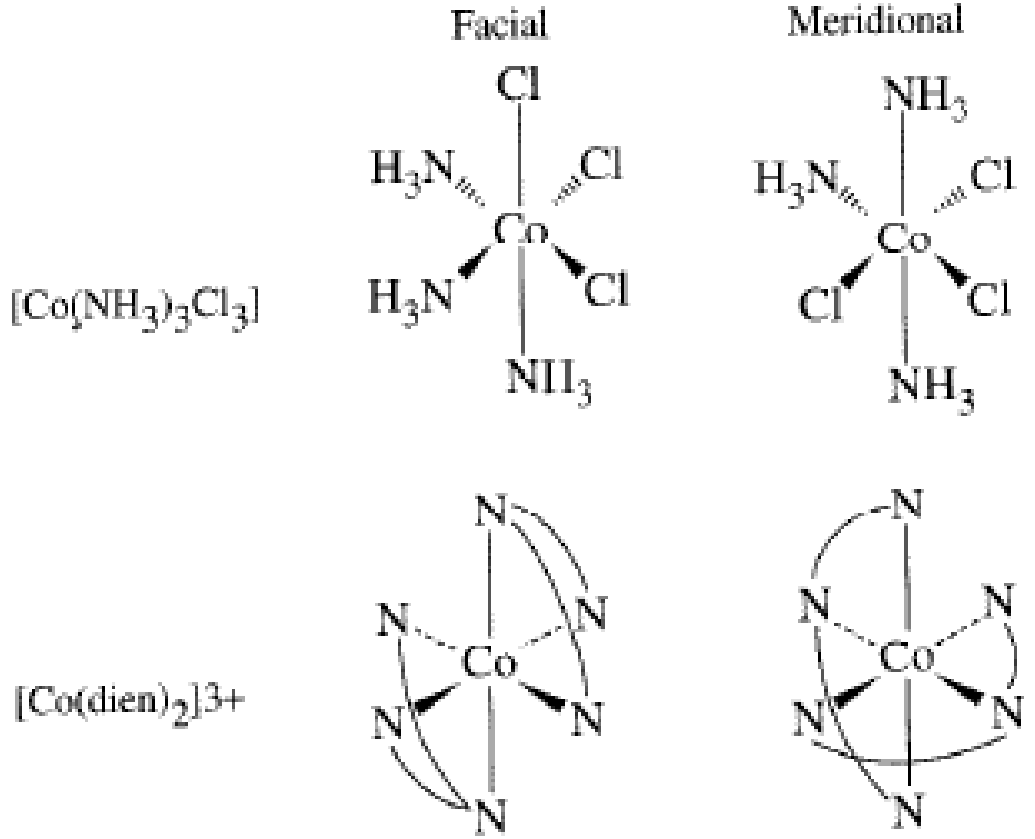


FIGURE 9-8 Facial and Meridional Isomers of $[Co(NH_3)_3Cl_3]$ and $[Co(dien)_2]^{3+}$.

tridentate	diethylenetriamine	2,2′-diaminodiethylamine or 1,4,7-triazaheptane	dien	$NH_2CH_2CH_2NHCH_2CH_2NH_2$

Special nomenclature has been proposed for other isomers of a similar type. For example, triethylenetetramine compounds have three forms: α, with all three chelate rings in different planes; β, with two of the rings coplanar; and *trans*, with all three rings coplanar, as in Figure 9-9. Additional isomeric forms are possible, some of which

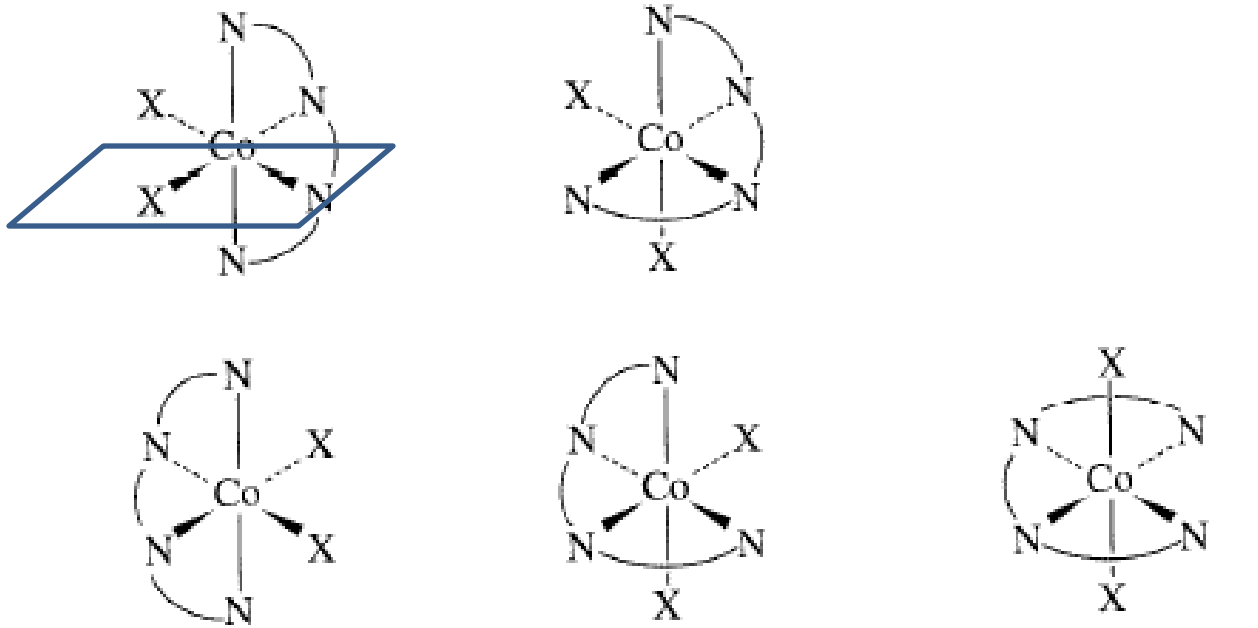


FIGURE 9-9 Isomers of Triethylenetetramine Complexes.

Hydrate isomerism

Hydrate isomerism is not common but deserves mention because it contributed to some of the confusion in describing coordination compounds before the Werner theory was generally accepted. It differs from other types of isomerism in having water as either a ligand or an added part of the crystal structure, as in the hydrates of sodium sulfate (Na_2SO_4, $Na_2SO_4 \cdot 7\ H_2O$, and $Na_2SO_4 \cdot 10\ H_2O$ are known). More strictly, it should be called solvent isomerism to allow for the possibility of ammonia or other ligands also used as solvents to participate in the structure, but many examples involve water.

The standard example is $CrCl_3 \cdot 6\ H_2O$, which can have three distinctly different crystalline compounds, now known as $[Cr(H_2O)_6]Cl_3$ (violet), $[CrCl(H_2O)_5]Cl_2 \cdot H_2O$ (blue-green), and $[CrCl_2(H_2O)_4]Cl \cdot 2\ H_2O$ (dark green). A fourth isomer, $[CrCl_3(H_2O)_3]$

6 coordinate complex

Ionization isomerism

Compounds with the same formula, but which give different ions in solution, exhibit ionization isomerization. The difference is in which ion is included as a ligand and which is present to balance the overall charge. Some examples are also hydrate isomers, such as the first one listed below.

$[Co(NH_3)_4(H_2O)Cl]Br_2$	and	$[Co(NH_3)_4Br_2]Cl{\cdot}H_2O$
$[Co(NH_3)_5SO_4]NO_3$	and	$[Co(NH_3)_5NO_3]SO_4$
$[Co(NH_3)_4(NO_2)Cl]Cl$	and	$[Co(NH_3)_4Cl_2]NO_2$

Coordination isomerism

4 coordinate complex

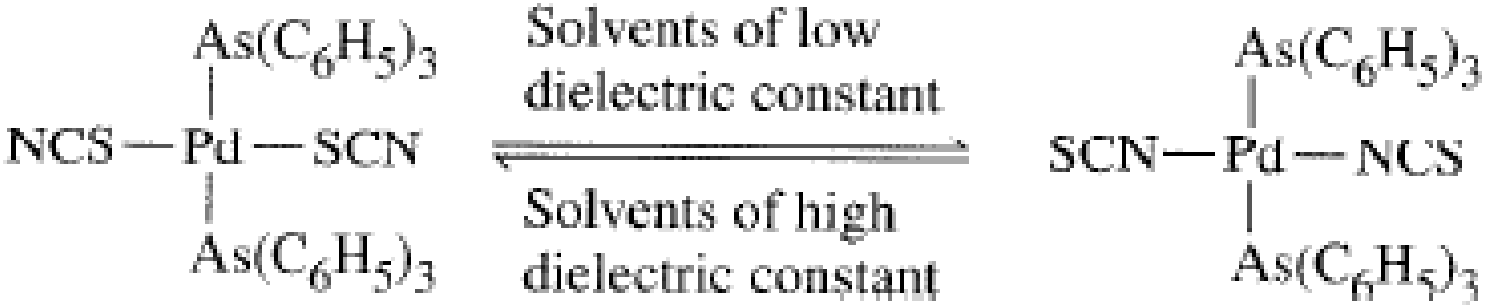
Examples of a complete series of coordination isomers require at least two metal ions and sometimes more. The total ratio of ligand to metal remains the same, but the ligands attached to a specific metal ion change. This is best described by example.

For the empirical formula $Pt(NH_3)_2Cl_2$, there are three possibilities:

$[Pt(NH_3)_2Cl_2]$	
$[Pt(NH_3)_3Cl][Pt(NH_3)Cl_3]$	(This compound apparently has not been reported, but the individual ions are known.)
$[Pt(NH_3)_4][PtCl_4]$	(Magnus' green salt, the first platinum ammine, was discovered in 1828.)

Linkage (ambidentate) isomerism 4 coordinate complex

Some ligands can bond to the metal through different atoms. The most common early examples were thiocyanate, SCN^-, and nitrite, NO_2^-. Class (a) metal ions (hard acids) tend to bond to the nitrogen of thiocyanate and class (b) metal ions (soft acids) bond through the sulfur, but the differences are small and the solvent used influences the bonding. Compounds of rhodium and iridium with the general formula $[M(PPh_3)_2(CO)(NCS)_2]$ form M—S bonds in solvents of large dielectric constant and



(a)

Due to a higher dentate character of ligands, a variety of complexes known as chelate is also formed sometimes. **Chelate** is a compound formed when a polydentate ligand uses more than one of its coordinating atoms to form a closed-ring structure, which includes the central metal ion. Five- and six-membered rings are known to provide extra stability to the chelates. The process of chelate formation is known as chelation. A polydentate ligand involved in chelate formation is also known as a chelating ligand. Chelates generally exhibit higher stability than analogous complexes.

5.4 Grinberg's Method

In support of Werner's coordination theory, A. A. Grinberg developed a unique experimental method to distinguish the *cis*- and *trans*-isomers in coordination complexes. It is an important chemical method in which a chelating ligand with two donor atoms is allowed to react with *cis*- and *trans*-isomers separately. With *cis*-isomers, both the donor atoms of a chelating ligand coordinate to the central atom, forming a chelate with a five- or six-membered ring. While in the case of the *trans*-isomer, the chelating ligand can coordinate to the central atom by only one of the donor atoms. Here it functions as a monodentate ligand and forms an ordinary complex but not chelate.

5.4 Grinberg's Method

In support of Werner's coordination theory, A. A. Grinberg developed a unique experimental method to distinguish the *cis*- and *trans*-isomers in coordination complexes. It is an important chemical method in which a chelating ligand with two donor atoms is allowed to react with *cis*- and *trans*-isomers separately. With *cis*-isomers, both the donor atoms of a chelating ligand coordinate to the central atom, forming a chelate with a five- or six-membered ring. While in the case of the *trans*-isomer, the chelating ligand can coordinate to the central atom by only one of the donor atoms. Here it functions as a monodentate ligand and forms an ordinary complex but not chelate.

Oxalic acid $(COOH)_2$, glycine ($H_2N—CH_2—COOH$) and ethylenediamine ($H_2N—CH_2—CH_2—NH_2$) are the most frequent chelating ligands used in this method. Reactions between *cis*- and *trans*-isomers of $[Pt(NH_3)_2Cl_2]$ and oxalic acid, as well as glycine ligands, are shown in Figure 36.

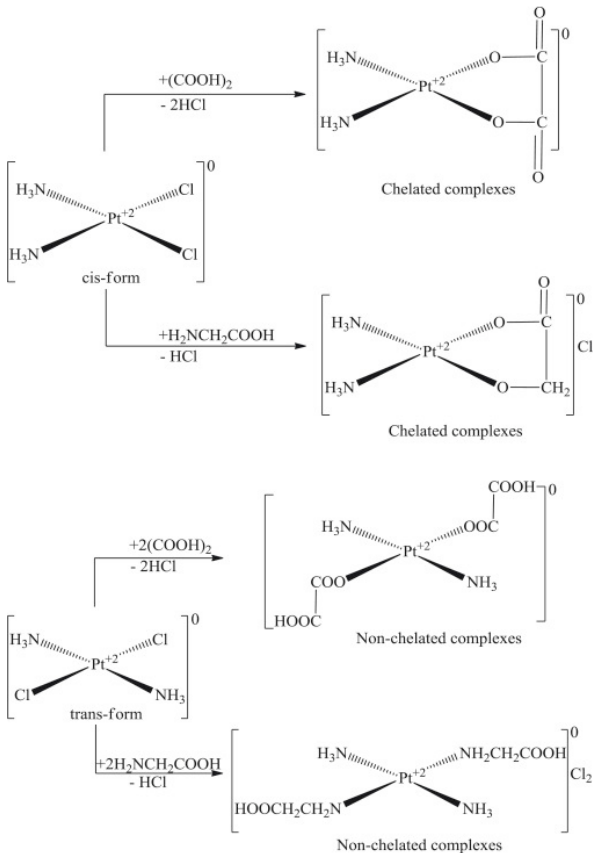
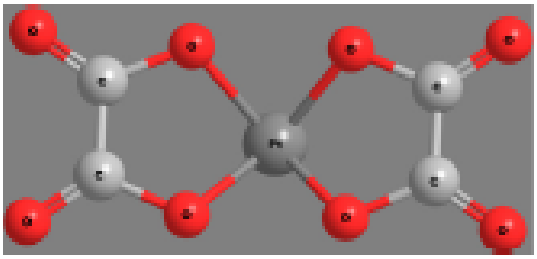
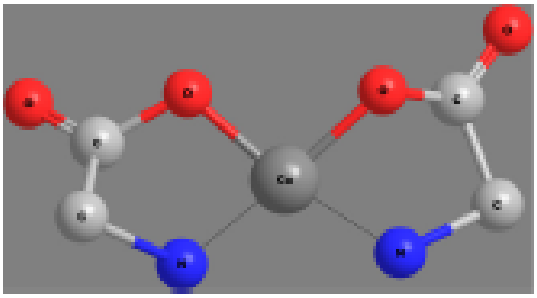
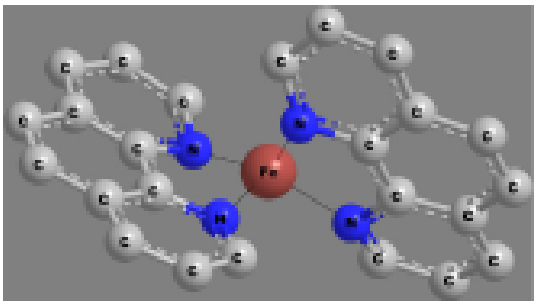
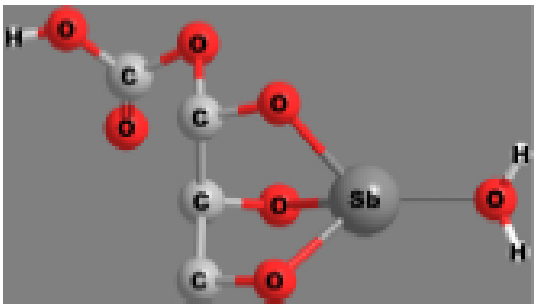
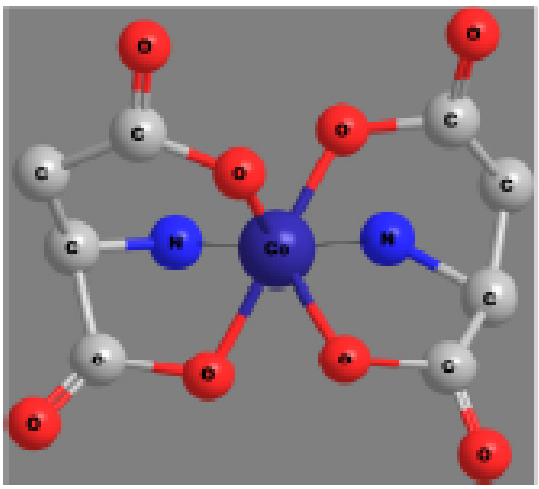


Table 2 Effect of some factors on the stability of complexes—cont'd

Sr.No.	Factor	Comments
7.	Basicity	Stronger Lewis bases readily donate electron pairs and form complexes with greater stability. In line of this, ammonia forms more stable complexes than water, which in turn forms more stable complexes than hydrofluoric acid.
8.	Chelate effect	Complexes containing chelate rings are generally more stable than those without rings. The value of stability constant for an ammonia complex with Ni(II) is 7.98, whereas with ethylenediamine (en) is 18.2. The differences in entropy between chelate and non-chelate complex reactions essentially cause this change in the values of stability constants. The formation of chelate results in greater disorder as it involves the formation of a larger number of free particles in the products, while there is no change in the number of particles during the formation of analogous non-chelated complexes. The chelates are formed when a polydentate ligand uses more than one of its coordinating atoms to form a ring involving the metal ion. Here, the dissociation of one of the bonds with the metal ion does not detach the ligand molecule, and hence, the possibility of reestablishment of this bond is more, as compared to an ordinary complex. Thus, chelation provides additional stability to a complex molecule.
9.	Number of chelate rings	Ethylenediamine (en) and diethylenetetramine (dien) are capable of forming one and two chelate rings, respectively. The 1:2 complexes of these ligands with Fe^{+2} ion register the values of stability constants of 7.7 and 10.4, respectively. The chelates containing five- and six-membered chelate rings show maximum stability.
10.	Steric hindrance	The Ni(II) complex of oxine (8-hydroxy quinolone) has higher stability constant value of 21.8, as compared to an analogous complex with 2-methyl-8-hydroxy quinolone (17.8)

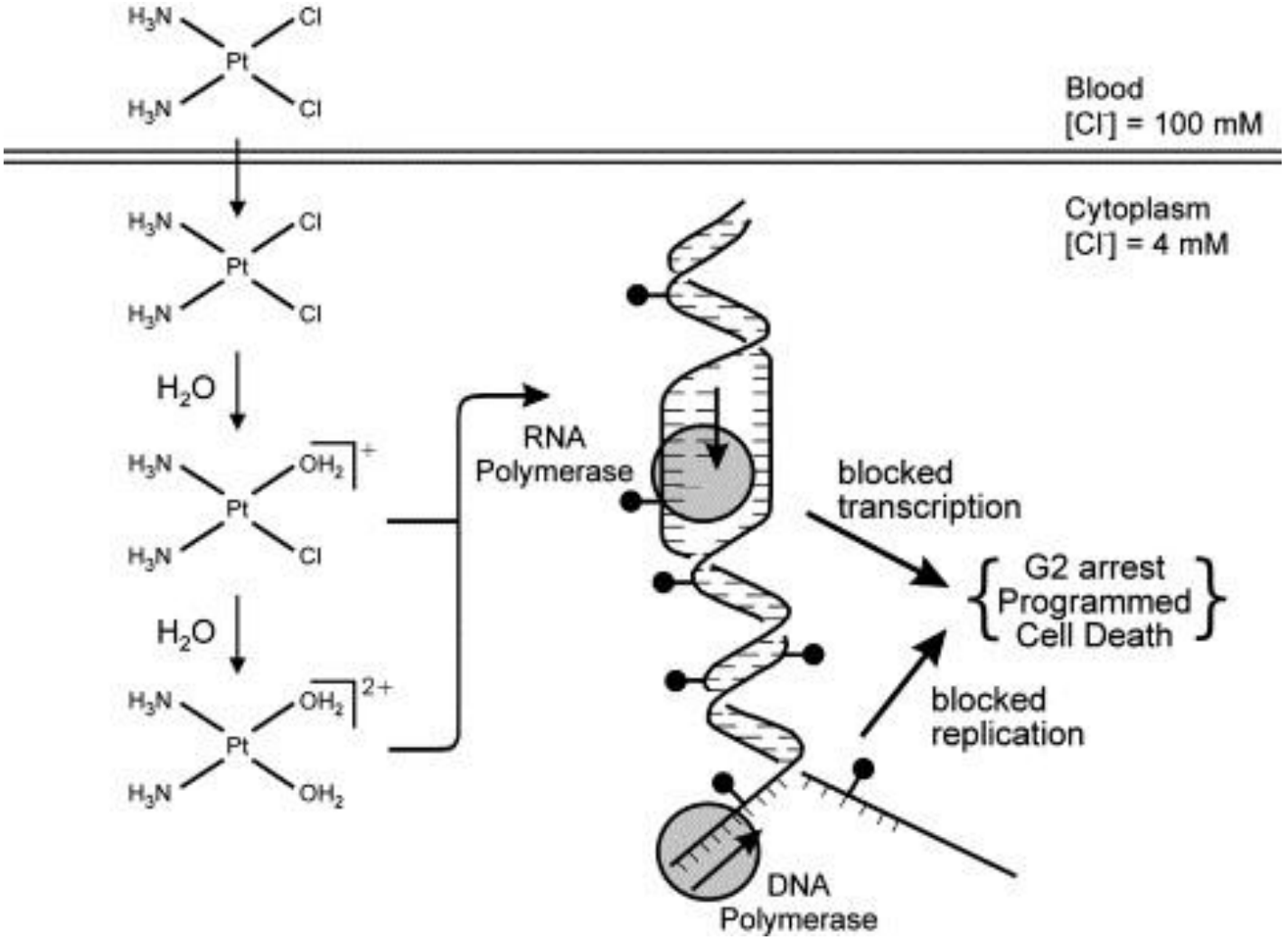
EXAMPLE	CHARACTERISTICS
	Oxalic acid Anionic Bidentate Two covalent bonds Two five-membered rings in the chelate
	Glycine Anionic Bidentate One covalent and one coordinate covalent bond Two five-membered rings in the chelate
	1,10-phenanthroline Neutral Bidentate Two coordinate covalent bonds Two five-membered rings in the chelate
	1,2,3-trihydroxypropyl hydrogen carbonate Anionic Tridentate Three covalent bonds Two five-membered rings and one six-membered ring in the chelate
	2-aminosuccinic acid also known as aspartic acid Anionic Tridentate Two covalent and one coordinate covalent bond Two five-membered rings, two six-membered rings and two seven memebered rings in the chelate

The chelates of this coordination number are less stable than analogous complexes. Consider the complexes $[Ag(en)]^+$ and $[(Ag(NH_3)_2]^+$. In these complexes, the favoured geometry is linear. In $[Ag(en)]^+$ chelate, a five-membered ring is formed, and the 'en' molecule cannot occupy linear coordination positions around Ag^+ without introducing strain in the chelate ring, which destabilizes the complex. Hence $[Ag(en)]^+$ is less stable than $[(Ag(NH_3)_2]^+$. The Ag(I) chelates with six-, seven- or eight-membered rings are more stable; as in larger-sized chelate rings, the strain is less, hence stability is greater.

Abstract

Coordination compounds and organometallics have very special place in chemistry due to their different structural arrangements and application in various fields. They also play vital role in biological system particularly in human and plant life. The blood oxygen carrier haemoglobin is a coordination compound of iron which is essential for human life, also the chlorophyll is a coordination compound of magnesium essential for plant the life and the continuation of the life on this planet. These compounds are not only biologically important but also comprise a group of drug like molecules and medicines. Coordination compounds such as platinum, palladium and ruthenium complexes are anti-cancer drugs. Also, many of the coordination compounds are anti-bacterial. The coordination compounds have been showed effective behaviour against many diseases like Alzheimer's and malaria. Recently coordination compounds used as antivirus such as coronavirus. In present review, we highlighted the most important coordination complexes that show a significant role in the field of medicine.

Anticancer: The difference between the cis isomer and trans of $[PtCl_2(NH_3)_2]$ is a geometry whereas *cis* isomer is biological activity and it uses as anticancer drug because the Cl ligand in *cis* isomer is active and it replaces by water molecule then the new product platinum complex with water reacts with DNA of the cancer cell by replacing water molecule which leads to inhibition growth the cells of cancer [9].



There are many drug coordination compounds based on Schiff bases. The complexes of copper (II) with Schiff bases of (6-methyl-2-oxo-1,2-dihydroquinoline-3-carboxaldehyde semicarbazones) and (8-methyl-2-oxo-1,2-dihydroquinoline-3-carboxaldehyde semicarbazones) (Figure 6) exhibited satisfactory cytotoxic activity against A431 cancer cell line.

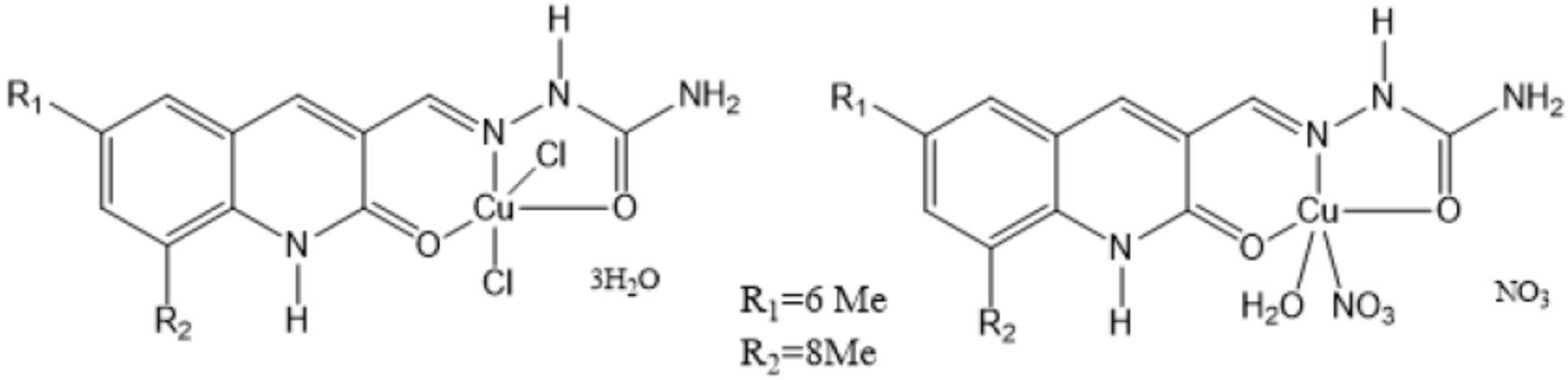


Figure 6 Structure of Cu(II) complexes having cytoxic activity against A431 cancer cell line

Antimalaria: The many metal complexes such as iron, gold, ruthenium, cobalt, rhodium, copper, cobalt, zinc, osmium and palladium etc. shown effective anti-malarial[15,16]. The complex of zinc (II) with amodiaquine (Figure 9) exhibited efficacy towards *Plasmodium berghei* with safety evaluation[17].

Other compound of ferrocenyl carbohydrate (Figure 12) shown potential as an anti-malarial compound[24–26].

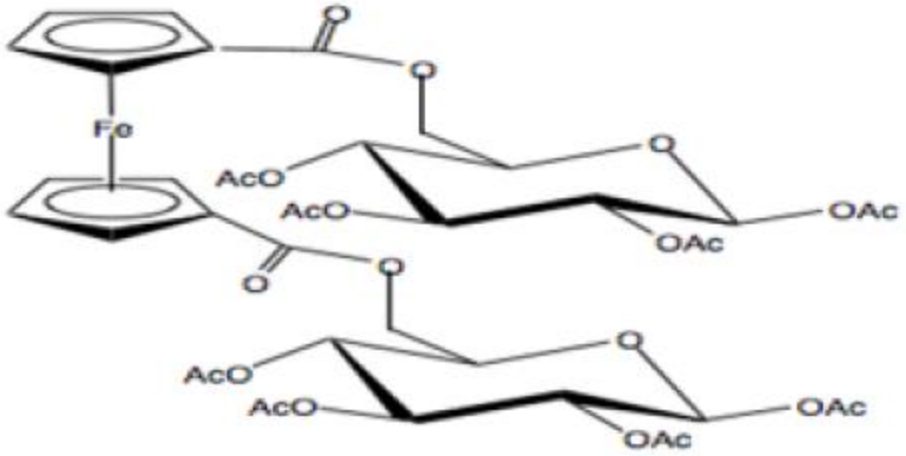


Figure 12 Ferrocenyl carbohydrate conjugate

Anti-Alzheimer: Metal complexes exhibited ability to blocking β-amyloid aggregation and scavenging its toxicity. Some complexes for ruthenium (III) exhibited significant role as anti-Alzheimer agents such as NAMI A, KP1019, and PMRU20 (Figure 14) [28]. In the same manner, some copper (II) and vanadium complexes exhibited remarks role in the *vitro* as anti-Alzheimer[29,30]

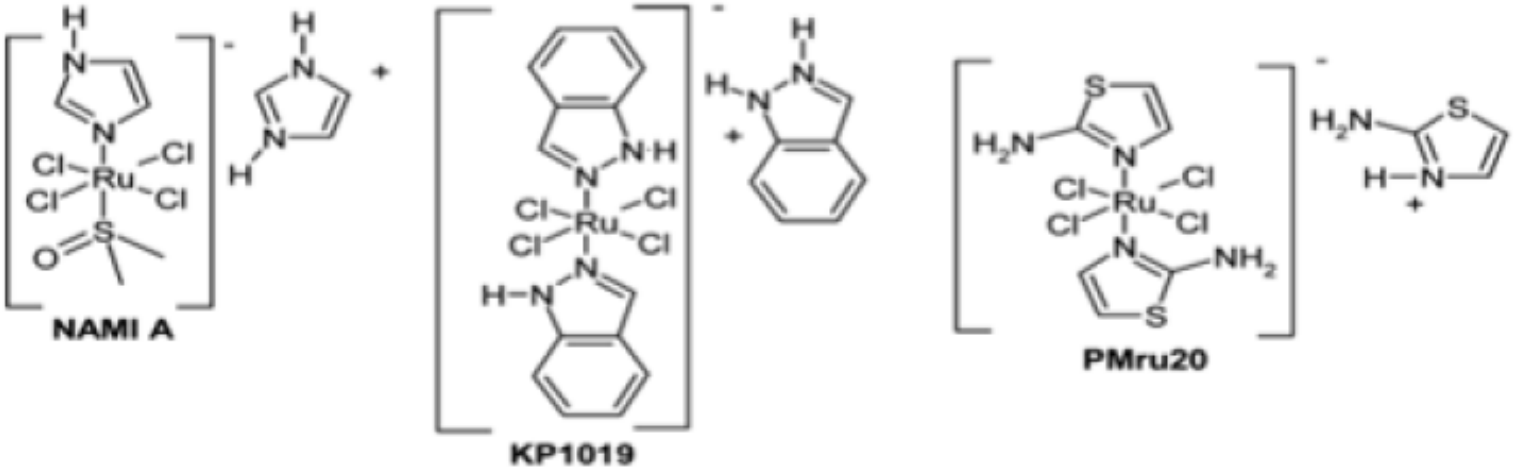


Figure 14 Ruthenium compounds as anti-Alzheimer